A National Airspace System Reference Architecture as a System of Systems

Tracing Operational Intent from Enterprise Objectives to Aircraft Trajectory

ISD 503 Capstone Project Report – Interim Report #2

Submitted to:
Mark Petrotta - Principal Engineer of Digital Engineering, MBSE, and Augmented Intelligence - Aerospace and Defense Systems
IOE Department of Integrated Systems Engineering & Design

Prepared by:
Owen O'Keefe
University of Michigan, ISD

October 19, 2026

Plan ref: to-do-list.md S1 — Confirm advisor/reader name and title for "Submitted to."

Executive Summary

Plan ref: to-do-list.md S14, S16 — Write this section LAST. Distill: (1) the improvement opportunity/problem statement, (2) the MBSE approach used, (3) major architecture and optimization-demonstration findings stated quantitatively, (4) the operational impact/benefit of the reference architecture (see Section 6), and (5) recommendations. Limit to 1-2 pages; no citations, undefined terms, or content not found elsewhere in the report. State the two claims here as well as in the Introduction (Section 1; owner's note on the 2026-10-11 print, page 11).

- **Problem.** NAS decisions are made by independent actors, each with partial information. When a conflict or disruption occurs, the decision is made quickly and myopically. The sector controller resolves conflicts knowing type, level, speed and route, but not weight or cost index.
- **Gap.** Few sources model the NAS with systems engineering. None found address what detailed aircraft state information is worth to the decision.
- **Claims.**
  - Experiment: a higher-fidelity decision aid, fed accurate performance data and the wider airspace state, changes how conflicts are resolved.
  - Model: an MBSE architecture locates the missing interface, names who owns the decision and the data, states the unmet requirement, and lets the scoring fidelity be swapped with the structure unchanged.
- **Approach.**
  - SysML v2 model of the NAS as a system of systems: context diagram, internal block diagrams, nominal-operations activity, requirements, verification cases, the proposed airline-to-ATC interface.
  - Experiment: one level-crossing conflict, four aircraft, one sector; four cases (A to D) with increasing information; options scored with OpenAP at three fidelities; impacts reported side by side for aircraft, airline, sector and system.
- **Findings: architecture.** [Measures from Section 5: requirements traced, requirements allocated, requirements tagged `improved`.]
- **Findings: experiment.** [Option chosen by each case; B minus A (value of automation) and C minus B (value of the information exchange) in fuel, CO2 and instructions; who pays.]
- **Impact.** [From Section 6: operational and dollar terms where a credible conversion exists.]
- **Recommendations.** [From the Conclusions and Recommendations section.]

Contents

Executive Summary **1**

1 Introduction **3**
1.1 Overview of the Project . . . 4
1.2 Literature Review . . . 5
1.3 Current State . . . 6
1.4 Stakeholders and Study Boundary . . . 7
1.5 Project Deliverables . . . 8
1.6 Project Timing . . . 8

2 Assumptions and Methodology **10**
2.1 Systems Engineering Approach . . . 10
2.2 Working Assumptions . . . 10

3 System Model **12**
3.1 Stakeholders . . . 12
3.2 Constituent Systems . . . 16
3.3 System of Interest . . . 16
3.4 Operational Concept . . . 19
3.5 Architecture: Detailed and Abstracted . . . 22
3.6 Proposed Airline-to-ATC Interface . . . 26

4 Experiment **26**
4.1 Scenario . . . 26
4.2 Cases and Resolution Options . . . 28
4.3 Scoring . . . 29
4.4 Aircraft Model and Its Simplifications . . . 31

5 Results and Discussion (Preliminary) **33**
5.1 Model Results . . . 33
5.2 Experiment Results . . . 33
5.3 Emissions and Who Pays . . . 34

6 Impact/Financial Benefits **35**

7 Conclusions and Recommendations **35**

A Appendices **43**
A.1 Full Nominal-Flight ConOps . . . 43
A.2 Full Stakeholder Register and RACCI Matrix . . . 49
A.3 Full SysML Diagram Set . . . 49
A.4 Full Myopic-Optimization Conflict Catalog . . . 53
A.5 Optimization Experiment Detail . . . 53
A.6 Generative AI Use Disclosure . . . 53
A.7 Source Register . . . 53

1 Introduction

Plan ref: to-do-list.md S1 — Briefly state the subject, purpose, and scope of the report: the NAS modeled as a System of Systems, centered on how operational intent propagates from enterprise objectives to aircraft trajectory/behavior.

This project builds a reference MBSE architecture for how airspace intent is managed across the National Airspace System (NAS). The architecture spans the organizations that define and enforce rules, the domains that manage airspace resources, the systems that support flight operations, and the onboard systems that turn intent into an executable trajectory and guidance behavior.

DeLaurentis [16] states the problem as follows:

> "Problems of significant complexity are facing decision-makers within government and industry, and the moniker of 'system-of-systems' (heretofore shortened to SoS) is increasingly being applied to problems of this type. Effective system analysis for decision-support quickly becomes unmanageable in this context, with multiple, heterogeneous, distributed systems involved and couched in networks at multiple levels. The situation is exacerbated by the fact that most organizations (and their processes) remain in the 'stovepipe' mode – their people and tools are configured to study within narrow bins with little or no analysis across those bins. Current frames of reference, thought processes, analysis, and design methods are not complete for these SoS problems ... A holistic framework is needed that enables decision makers to discern whether related infrastructure, policy, and/or technology considerations together are good, bad (or indifferent) over time."

The statement dates from 2005, but still describes the NAS today. The NAS is a system of systems, and the decisions its stakeholders make have consequences for the safety, efficiency, and sustainability of air transportation.

Airlines invest significant time and money in operational plans that maximize revenue. On the day of operation, unpredictable events – bad weather, aircraft malfunctions, crew absenteeism – can substantially alter that plan and the revenue objectives behind it [11]. To manage this, airlines maintain an Airline Operations Control Center (AOCC), a department of specialists in disruption management across aircraft, crew, and passenger recovery. The AOCC monitors operations and makes real-time decisions – re-routing flights, rescheduling crews, rebooking passengers – intended to limit the effect of a disruption on the airline's schedule and revenue.

The literature does not converge on one standard definition of the "cost" the AOCC is minimizing, and the values it does quantify do not always hold up under scrutiny. The optimization literature commonly models delay cost as a linear function of delay minutes, but Hu et al. [44] note that this is inconsistent with airline practice, where delay cost is a nonlinear function of delay time, and that, faced with this complexity, airlines tend to put the speed of a recovery decision ahead of the mathematical optimality of the solution. The same preference for fast, tractable decisions shows up in how the work is structured. A review of the Aircraft Recovery Problem finds that studies – and, by extension, the AOCC processes they model – rarely minimize delay, cancellations, and aircraft swaps together. They treat aircraft, crew, and passenger recovery as separable sub-problems, solved in sequence and handed off in turn [77]. Each stage can produce a solution that is locally optimal yet globally suboptimal, or even infeasible, once the next stage's constraints are applied. This is a documented instance of the myopic-optimization pattern this capstone investigates.

The pattern is visible even in work that improves on the status quo. The predictive turnaround-buffer allocation model of Kontodimou et al. [56] achieves a 37.1% reduction in propagated delay

relative to a zero-buffer baseline, but it reaches that result by holding aircraft routing, fleet assignment, crew duty, maintenance, and passenger connectivity fixed as upstream constraints, outside the optimization. The gain is real within that scope. The framing also shows the risk this capstone is concerned with: a locally optimized decision layer can report a strong local benefit while passing tradeoffs to adjacent decision layers that its own cost model does not measure. The same pattern appears on the human side of the AOCC. Turhan, Gzara, and Elhedhli [89] document that when dispatcher workload exceeds capacity, dispatchers "may resort to conservative approaches, which can have financial implications for airlines" – a workload-driven cost that rarely appears in the delay, cancellation, and swap cost models discussed above. Hassan, Santos, and Vink [36] frame this more broadly as a persistent gap between the disruption-management optimization literature and AOCC practice.

This report explores the value tradeoffs aviation stakeholders experience when a disruption occurs and the recovery decision is made with more or less myopia. A dispatcher or AOCC team member making a decision is often focused on its immediate effect – for example, minimizing delay for one flight – without visibility into what it means for the airline's wider operation or for the NAS as a whole. The cost models in use today are over simplifications that capture the factors easiest to quantify, such as fuel and crew cost. Factors that are harder to quantify – maintenance cost, passenger satisfaction, dispatcher and controller workload, environmental impact – are more often left out. The result can be congestion in particular airspace sectors, missed opportunities for more efficient routing, or lower profit driven by customer dissatisfaction. By modeling the NAS as a system of systems and tracing how operational intent propagates through it, the project aims to identify ways to improve these decisions and reduce the cost of disruptions to the air transportation system.

Two claims are central to the project. The first, explored by the experiment, is that replacing a low-fidelity rule of thumb with a higher-fidelity model fed by accurate performance models and informed of the broader state of the airspace can and will lead to changes in how disruptions are mitigated. The second, shown by the model, is that an MBSE architecture can locate that interface, name who owns the decision and the data, state the unmet requirement, and lets the scoring model's fidelity be swapped (conceptual, nominal mass, actual mass) with the structure unchanged. As such, the model contributes is a traceable representation of how strategic objectives become executable aircraft behavior.

1.1 Overview of the Project

In the NAS, operational decisions are made by independent actors, each of which may or may not account for the effect on the others. Those actors include regulatory agencies, air traffic service providers, airport operators, airline operations centers, maintenance (MRO) suppliers, flight crews, passengers, and the aircraft systems themselves. Substantial architectural work exists within individual domains, but the relationships and information flows between domains are often incompletely represented.

Existing models frequently focus on one actor or one problem, and show one or more of the following limitations:

- emphasis on enterprise-level capabilities without sufficient representation of aircraft behavior and onboard decision-making;
- emphasis on aircraft systems without adequate representation of the operational context in which those systems operate;

- limited traceability between strategic objectives, operational constraints, human decisions, and the resulting aircraft trajectories;
- difficulty evaluating the effect of new operational concepts, automation technologies, and optimization services across organizational boundaries.

As a result, no broadly applicable architecture fully captures how strategic objectives become executable aircraft behavior. Without end-to-end traceability, engineers, operators, and decision makers have limited means to understand the consequences of a change within the NAS, and integrating future decision-support technology is harder than it needs to be.

This project addresses that gap by developing a System-of-Systems architecture that explicitly models the propagation of intent through the NAS, from enterprise objectives and airspace management decisions to flight deck decision-making and aircraft trajectory execution.

1.2 Literature Review

A literature review explored the makeup of the NAS more generally and the various problem spaces that it contains. The identified literature can be categorically organized as follows:

- Long-horizon strategic airline planning and operations
- Day-of-operations and collaborations necessary to begin a flight's execution
- Tactical flight execution and disruption management

Airline planning and operations literature focused on schedule planning, fleet assignment, crew scheduling, and maintenance routing. In general these operations consist of a series of optimization problems that are solved sequentially, long before a flight enters the NAS, and the chosen solutions define the input conditions to the day-of-operations and turnaround, and OCC/dispatch and flight execution problem spaces, which are more directly relevant to the intent-to-trajectory problem space this capstone addresses.

The literature focused on day-of operations explored how airlines and airports coordinate to manage flight schedules, gate assignments, and turnaround times, as well as how weight and balance considerations affect aircraft performance and safety. The OCC/dispatch and flight execution literature focused on disruption management, dispatcher responsibilities, and nominal ATC/IFR flow.

Very few of the sources utilized a systems engineering approach to model the problem space and the relationships between the various actors and systems. [58] developed a model of the NAS using MBSE to represent the various actors and systems, but it did not focus on the propagation of intent through the system and did not publish their models publicly.

The EU Greener Air Traffic Operations (GREAT) project [43] developed a model of the NAS using MBSE to represent the various actors and systems, which was the most comprehensive found. On page 37 the authors explore the modifications made to a flight throughout the cruise phase and how those are communicated and coordinated between the various actors and systems. However, the GREAT project did not address the value of detailed state information and the effect of that information on decision-making and trajectory execution.

The lack of modelling is a gap in the existing research, which this capstone addresses by developing a System-of-Systems architecture that explicitly models the propagation of intent through the NAS, from enterprise objectives and airspace management decisions to flight deck decision-making and aircraft trajectory execution.

The gap represented by the lack of information exchange and the effect of that information on decision-making and trajectory execution is addressed by the experiment, which evaluates the effect of providing additional information to the ATC sector controller on the resulting trajectory and its associated costs. [15] measured what airline-supplied weight does for climb prediction. Much work has explored the potential value of EPP, which will provide for downlink of the FMS's predicted profile, but it has not been widely implemented and it does not provide for what-iffing.

Furthermore, the experiment explores the impact of how the controller defines optimality when resolving a conflict between two aircraft. This project bridges these gaps by:

- scoring one decision at four stakeholder levels side by side (aircraft, airline, sector, system) with no weighted sum, so who pays is visible;
- mapping the decision to its authority and its information owner in a system-of-systems model, so the missing exchange is a named interface and an unmet requirement, not a finding in prose;
- swapping the scoring model's fidelity under a fixed architecture and measuring its impact, which is the MBSE claim.

In the moments when the conflicts occur and disruption managment must occur, the controller has type, level, speed and route, but not weight or cost index. [79] describes how weight and speed intent "are considered competitive parameters by airlines and are not available for use". As such, the what-iffing aspect of the additional interfaces must be executed carefully and with security in mind.

[73] describes how this lack of information leads to simplified solutions, where controllers "use more predetermined rules", leading to 23 of 36 level-crossing cases resolved by a descent. This simplification is further supported by the findings of [55], whos investigations determined that controllers "leave the aircraft in steady state alone, and by [34] who quoted one controller as stating there's "No point asking aircraft to climb if they come back and say they can't".

[19] explored the value of providing traffic information to the flight deck where it can be combined with performance data to determine optimal trajectories, and found that it led to a reduction in fuel burn and time for the typical case. This project is a mirror-image study, where the traffic picture is provided to the controller instead of the flight deck, and the effect on the trajectory and its associated costs are measured. The study did not model aircraft weight and limited climbs to FL350 and below for that reason. This is exactly the assumption Section 4.2 adopts for a point of comparison.

1.3 Current State

For a period the project lacked a specific focus for which tradeoffs it would consider. The project is not an exploration of optimization, but its concept centers on sub-optimality driven by myopic decisions. The concepts of operations gave a more complete understanding of the NAS, the systems involved, and the interfaces among them, but they did not by themselves point to a specific scenario worth modelling.

The pivot from Phase 1 exploration to Phase 2 modelling was made to advance the project deliverables while building a deeper understanding of the information gaps. The intent was that when the modelling reached the measures of effectiveness and performance, the phase 1 content would be revisited updated as warranted.

Building the model raised questions the literature review alone had not: who owns the decision support for a trajectory change, which party holds the information that decision needs, and whether an exchange between them exists. Answering them located one specific gap. The sector controller

decides which aircraft to move in a conflict, the airline and the flight management system hold the aircraft's weight and cost index, and no exchange carries one to the other. That gap is now a named interface in the model (`AirlinePerformanceShare`) and an unmet requirement. The literature consultation that followed was targeted in the way the pivot intended: sources on controller conflict-resolution practice [73, 55, 34], on climb prediction with airline-supplied weight [15], and on traffic-aware flight-deck requests [19] were found and read because the model and the experiment needed them.

A 2014 study of traffic-aware requests [19] found that providing the flight deck with traffic information led to a reduction in fuel burn and time for the typical case, enabling savings over $3.3M annually. The study did not model aircraft weight and used conservative estimates of aircraft performance, which lead them to avoid climbs for aircraft above FL350 for that reason. Mori [69] starts from the same absence and posits why aircraft mass is usually not openly available - because airlines treat it as confidential.

The project is now organized as four parallel tracks - model, experiment, report, and on-demand evidence - which are being worked in nine one-week turns that end on 2026-12-13. This plan replaced the single-pass August list on 2026-10-10, which is statused below. One throughline carries the project: the airline's cost objective, the stakeholder and system requirements that follow from it, the airline-to-ATC interface, the conflict-resolution decision, and the experiment that scores that decision.

The requirements have been drafted and revised through two hand reviews and now stand at 13 stakeholder requirements and 23 system requirements, each limited to what bears on the studied decision. A third tier of 28 constituent-system requirements is drafted. The model holds the context diagram, and several draft diagrams, including the aircraft internal block diagram, an airspace management internal block diagram, and nominal-flight activity diagram, and the typed airline-to-ATC interface. The state machines the original plan scheduled for 9/15 have not been built.

The experiment is defined and runs end to end. Four aircraft in one generic, overly simplified en-route sector present one crossing conflict, and four cases resolve it with different information: a controller's rule, automation with one assumed nominal weight per aircraft type, automation with the shared weight and cost index, and the same with passenger connection information. An open aircraft performance model, OpenAP, scores every option. The first results are in Section 5.2 and are preliminary.

1.4 Stakeholders and Study Boundary

Informed by the literature survey and 16 years of experience in the aviation industry, a broad set of candidate stakeholders was identified with a PESTLE analysis. Four are kept: operators (the airline and its operations center), air traffic control, the airport authority, and the flight crew. The ommitted stakeholders supply rules, equipment, demand, and expectations to the decision but do not make it or execute it. As such, the governments and standards organizations, aircraft and engine manufacturers, passengers, and environmental and societal organizations are not explicitly modelled. Section 3 gives the full stakeholder analysis (Section 3.1), the reason for each choice (Section 3.1.3), and the resulting System of Interest (Section 3.3).

The study examines one tactical decision, the resolution of a conflict between aircraft in cruise, and keeps only the stakeholders and objectives that bear on it. Six objectives are kept: safety, efficiency, environmental impact, schedule reliability, cost, and workload. Throughput and capacity are left for another study, and the phases before departure enter only as the conditions the studied decision inherits.

1.5 Project Deliverables

The project delivers the eight items of Table 1.

Table 1: Project deliverables.

Deliverable	What it shows	Where in this report
Context diagram	The systems kept in the System of Interest, the actors at its boundary, and the exchanges among them	Section 3.3
Block definition diagrams	The NAS broken down into domains and constituent systems	Appendix A.3
Internal block diagrams	The parts inside the aircraft and inside airspace management, and the links among them	Section 3.5
Activity diagrams	One nominal flight from gate to gate, and the place of the conflict within it	Section 3.4
Requirements	Stakeholder requirements, system requirements and constituent-system requirements, each traced to the tier above	Appendix A.3
Experiment	One conflict resolved four ways and scored for the aircraft, the airline, the sector and the system	Sections 4 and 5.2
Traceability chain	One thread from the airline's cost objective through the requirements and the airline-to-ATC interface to the experiment's outcome table	Section 5.2; Appendix A.3
Gap analysis	The requirements that today's NAS does not meet, and what the experiment measures for each	Section 5

1.6 Project Timing

The plan established at the project's inception is presented in Table 2 for reference, with the status of each item: not started, draft, complete, or will not implement.

Table 2: Project milestones and verification methods of the original plan, with the status of each.

Project Milestone	**Deliverable**	**Lead/Team Member**	**Expected Completion Date**	**Target Metric/Verification Method**	**Status**
Literature Review	Annotated bibliography	Owen	9/7/2026	10–20 papers categorized	Draft
PESTLE Analysis & Systems-of-Interest Definition	Context diagram, text-based documentation	Owen	9/10/2026	Review	Complete
Systems-of-Interest Definition	Text-based documentation	Owen	9/15/2026	Clear boundaries and exclusions	Complete
Operational Scenarios	ConOps document	Owen	9/22/2026	1–3 representative scenarios	Draft
Draft Architecture	BDDs and package structure	Owen	10/1/2026	All domains represented	Complete
Interface Definition	Interface Control Document	Owen	10/12/2026	All interfaces identified	Draft
Internal Block Diagrams	IBDs	Owen	10/17/2026	Critical interfaces connected	Draft
Requirements Definition	Requirements model	Owen	10/20/2026	Trace to stakeholders	Draft
Measures of Effectiveness	MOE/MOP hierarchy	Owen	10/25/2026	Quantifiable metrics defined	Complete
Requirement Traceability	Requirements matrix	Owen	10/29/2026	Complete allocation	Draft
Activity Diagrams	Operational activities	Owen	11/3/2026	Scenario walkthrough	Draft
Sequence Diagrams	Information exchanges	Owen	11/8/2026	Message completeness	Not started
State Machines	State behavior of key entities	Owen	11/13/2026	Executable state logic	Will not implement
Authority & Responsibility Models	RACII (Responsible, Accountable, Consulted, Informed)	Owen	11/17/2026	All decisions assigned	Will not implement
Optimization Prototype	Working code	Owen	11/21/2026	Unit test cases	Draft
Integration into MBSE Architecture	Traceability links	Owen	11/24/2026	End-to-end scenario	
Final Baseline & Final Paper	Versioned model & thesis/paper	Owen	12/16/2026	Draft	Not started
Paper Submittal	Thesis/paper	Owen	12/20/2026	Official submittal	Not started

2 Assumptions and Methodology

The problem is that the inertia of past, piecemeal designs and the high cost of upgrading the NAS make the system hard to change. The project proposes a new airline-to-ATC interface and a new way of deciding which aircraft to move in a conflict, which together could lower the cost of air traffic management.

Toward those ends, the project uses a model-based systems engineering (MBSE) approach to define the architecture and the decision-making process, and a simulation-based experiment to test the proposed updates to them. First, a literature review was conducted to:

1. identify the current state of air traffic management,
2. identify the relevant stakeholders and their objectives,
3. identify the requirements necessary to meet those objectives, and
4. identify the gaps in the current system that the proposed interface and decision-making process can address.

The contextual information captured in the literature review enabled the execution of the systems engineering approach.

2.1 Systems Engineering Approach

Standard systems engineering methods of analysis and decomposition were applied to the collected evidence, which produced a context diagram, internal block diagrams, and activity and sequence views. These identified the relevant systems and the interactions among them that are necessary to meet the stakeholders' objectives. From them the stakeholder requirements were written and then decomposed into system-level requirements (SLRs) for the NAS. The SLRs were allocated to the relevant constituent systems, and high-level requirements (HLRs) decomposed them one level further.

This enabled the project to (1) build a model of the architecture and the decision-making process, and (2) compare current air traffic management practice, which relies on general rules of thumb and knowledge of what has worked before [73, 55], with a model-based decision-support approach that replaces the low-fidelity decision rule with one or more higher-fidelity models of aircraft performance, at varying levels of accuracy and information.

The emphasis is on structure, behavior, interfaces, and traceability, not on optimization mathematics. The project's commercial value to the author lies in a better understanding of the systems, their interactions, and the divisions of responsibility that inhibit information flow and lead to suboptimal outcomes. To explore that in a meaningful way, decision support and optimization are treated as a capability inside the architecture, which following MBSE principles makes possible. No alternative system decomposition was formally explored beyond the initial abstraction decisions. The ISO/IEC/IEEE 15288 technical processes were followed in this tailored form, which the standard provides for.

Table 3 summarizes the systems engineering processes and the artifacts that realize them.

2.2 Working Assumptions

The working assumptions fall into two groups: assumptions about the problem and its context, and assumptions about the methodology.

Table 3: Systems engineering processes and the artifacts that realize them.

Process (15288 / INCOSE)	Artifact in this project	Where
Stakeholder needs and requirements	13 stakeholder requirements, each traced to an objective row	`requirements_stakeholders.sysml`; Appendix A.3
System requirements	23 system requirements, allocated to constituent systems; 28 constituent-system requirements one tier down	`requirements_nas_system.sysml`; `requirements_constituent_systems.sysml`
Architecture definition	Context diagram, internal block diagrams, typed interfaces on the thread, activity and sequence views	`cameo_models/`; Section 3.5
System analysis	Experiment at three fidelities, outcome table	`simulation/`; Sections 4 and 5.2
Verification	Verification cases for the improved requirements, run by the experiment	`nas_verification.sysml`
Validation	Stakeholder-level reading of the outcome table (who pays) with MoEs or MoPs	Section 5

The problem statement is that the NAS is hard to change, and that a new airline-to-ATC interface and a new way of deciding which aircraft to move in a conflict could lower the cost of air traffic management. This framing assumes that:

- the abstractions used in the model are sufficient to capture the relevant behavior of the system;
- the stakeholders' objectives are comprehensively captured and accurately represented in the model;
- the scenario is sufficient to explore the potential benefit of interconnected MBSE.

Every abstraction was made with the express goal of capturing the relevant behavior of the system, as defined by the stakeholders' objectives. The System of Interest, the NAS as a system of systems, was ratified around the trajectory-intent lifecycle, from enterprise objectives through aircraft motion. An entity is modeled when it has distinct decision authority or execution behavior that is dynamically coupled to in-scope decisions and materially affects cost, safety, workload, schedule reliability, or passenger value. Boundary actors supply objectives, demand, feedback, or rules without being decomposed as decision mechanisms. Exogenous influences enter as constraints or scenario parameters. This criterion preserves the authority and responsibility relationships a system-of-systems architecture needs without expanding the project into a model of every NAS participant.

The methodology is a model-based systems engineering approach that uses simulation to test the proposed interface and decision-making process. It assumes that:

- models of every required aircraft type are available;
- the security concerns of sharing airline data can be mitigated;
- the proposed interface can be implemented in the NAS over digital communication channels between the airline and ATC, between the aircraft and ATC, or both;
- the airline reports its weight and cost index accurately;

- the conflict can be treated as a snapshot, with the traffic unchanged while the decision is made;
- the specific scenario presented in the experiment is representative of a larger set of scenarios that generalize to the NAS as a whole.
- wake turbulence is not modeled, so the aircraft are assumed to be separated by more than the minimum wake-separation distance.

The explored scenario stands for a narrow set of conflicts: those between two aircraft crossing at the same level. Rantanen and Wickens [73] found that this represented only 36 of 256 conflicts identified in the operational data they studied. The scenario is represented through one hypothetical en-route sector and four aircraft, with controllers responsible for maintaining separation and for resolving conflicts. The aircraft are assumed to be at the same altitude, with the same speed, and on a collision course. The scenario is designed to be simple enough to allow for a clear analysis of the proposed interface and decision-making process, while still being representative of the types of conflicts that occur in the NAS.

The controllers are assumed to have accurate information about the aircraft and their intentions, and are equipped with varying levels of information and automation. In the baseline scenario, the controllers have only the information that is currently available to them in the NAS. In the proposed scenario, the controllers have access to additional information about the aircraft's weight and cost index, which allows them to make more informed decisions about which aircraft to move in a conflict. Inherent in this assumption is that the reported state information is accurate ([57] discusses how airlines can game the system with inaccurate information) and that the underlying performance models are sufficiently accurate to support the decision-making process (its simplifications are listed separately in Section 4.4). The controllers are also assumed to have the ability to communicate with the aircraft and the airline, and to be able to issue instructions to the aircraft in a timely manner.

3 System Model

This section presents the model the experiment is built on, beginning with the stakeholders, their objectives and the conflict among them. It then identifies the constituent systems through which they act, and draws the boundary of the System of Interest. It then describes how a flight operates within that boundary, the architecture that carries it, and the airline-to-ATC interface whose use informs the decisions that determine the value achieved by the experiment of Section 4.

3.1 Stakeholders

3.1.1 Identification

Informed by the literature survey and the author's experience in the aviation industry, a set of candidate stakeholders was identified and refined using a PESTLE analysis and stakeholder discovery process. The result is a diverse group of stakeholders that represent the various interests and perspectives affected by NAS operational decisions, and is shown in Figures 1

The stakeholders are grouped by the six domains of a PESTLE analysis (political, economic, social, technological, legal, environmental) and scored for importance and impact on trajectory decisions in 2.

The assessment is documented in the `capstone_stakeholder_assessment.xlsx` file, along with an initial assessment of each candidate stakeholder's objectives, candidate MOPs and MOEs, and

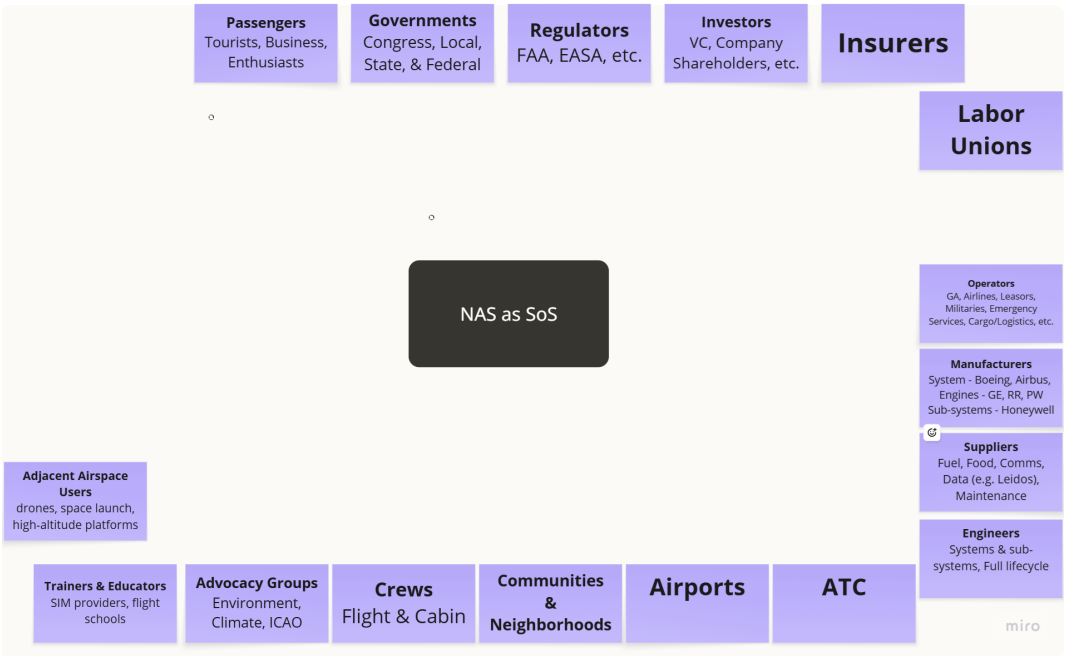


Figure 1: Candidate stakeholders of the NAS as a system of systems.

whether the stakeholder has a direct or indirect effect on trajectory decisions. The assessment also considers whether the stakeholder should stand alone or be abstracted through consolidation with another, and the reason for that decision. An excerpt of the work is in Appendix A.2, Table 7.

The power analysis informed the decision to keep four stakeholders in scope: the airline, air traffic control, the airport authority, and the flight crew. The other stakeholders are treated as boundary actors that supply rules, equipment, demand, and expectations to the decision but do not make it or execute it. The four kept stakeholders appear on the context diagram (Figure 4) and are decomposed in the model.

3.1.2 Candidate Objectives

An exploration of each stakeholder's interests led to the following list of objectives relevant to the project:

Each item is an objective because at least one stakeholder seeks to affect it, and it can be measured, as a measure of performance for one system or a measure of effectiveness for the whole.

Safety / Security Keeping aircraft separated from each other, from terrain and from hazardous weather, and protecting the operation from deliberate harm. Held by every stakeholder; for air traffic control and the flight crew it is a constraint that is never traded.

Efficiency Flying the flight with the least fuel and time for the distance covered. Held by the airline, whose fuel cost depends directly on the trajectory flown and whose time cost is determined by the passengers and the number and seniority of the crew conducting the flight.

Throughput / Capacity The number of aircraft a sector, runway or airport can handle in a period. Held by air traffic control and the airport authority.

Reliability / Resilience The ability of the operation to run as planned and to recover when it is disrupted. Held by the airline, air traffic control, and passengers.

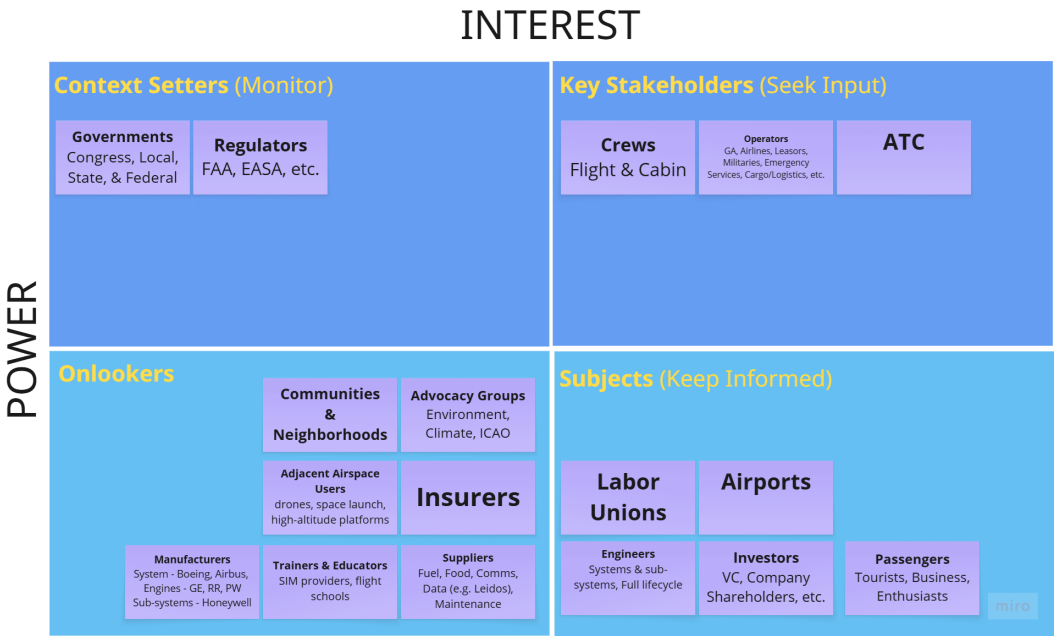


Figure 2: Candidate stakeholders placed by power and interest.

Environmental Impact The carbon dioxide, other emissions and contrails a flight produces, which follow from the fuel it burns and the path it flies. Held by environmental and societal organizations and, through regulation, by governments.

Noise Impact Sound over the communities near airports, set by the departure and approach paths. Held by communities and the airport authority.

Passenger Satisfaction The passenger's experience of travel time, comfort and connections. Held by passengers, and by the airline through the revenue that follows.

Schedule Reliability Arriving on time and keeping passengers' connections. Held by the airline and its passengers.

Cost / Profitability The operator's cost of a flight (fuel, crew, the consequences of delay) and the profit it supports. Held by the airline and its investors.

Maintenance / Lifecycle Cost The upkeep of aircraft and engines over their life. Held by the operator and the manufacturers.

Regulatory Compliance Operating within the rules, procedures and clearances in force. Held by the flight crew and the airline, and enforced by the regulators.

Workload / Human Factors The mental and procedural demand on controllers, dispatchers and crews. Held by air traffic control and the flight crew, for whom high workload is a cost and a source of error.

These objectives were then assessed for relevance to the decision under study, and six were kept: safety, efficiency, environmental impact, schedule reliability, cost, and workload. The other objectives were set aside for the reasons given in Section 3.1.3.

3.1.3 Selection

The stakeholders and objectives above are the result of exploration. The study itself is narrower. It examines one tactical decision, the resolution of a conflict between aircraft in cruise, and keeps only the stakeholders and objectives that bear on that decision.

Four stakeholders remain, together with the aircraft, and each appears on the context diagram (Figure 4): operators, as flight operations and its airline operations center; air traffic control, within airspace management; the airport authority, as airport operations; and the flight crew. The aircraft appears as aircraft systems. Governments and standards organizations, aircraft and engine manufacturers, passengers, and environmental and societal organizations are outside the System of Interest. They supply rules, equipment, demand, and expectations to the decision but do not make it or execute it, so they are treated as boundary actors and not decomposed.

Six stakeholder objectives remain: safety, efficiency, environmental impact, schedule reliability, cost, and workload. Schedule reliability is carried in the experiment as a step in cost when a flight's delay passes a threshold (the parameter K of Section 4.3). The others were set aside for the following reasons:

- Noise impact: the measure is "not relevant at cruise."
- Passenger satisfaction: "not measurable objectively."
- Profitability falls out of the other objectives; it is tracked at the operator's organizational level and is more complex than any one decision.
- Maintenance and lifecycle cost are largely determined by the number of flight cycles, the chosen takeoff performance, the flight distance and the climb controls, so one air traffic control decision has a negligible effect on them.
- Security, resilience and regulatory compliance are not changed by the choice between resolution options in the scenario.
- Throughput and capacity were left for another study. They are also close to ongoing work by the author, and out of an abundance of caution that part of the problem space was avoided.
-

3.1.4 Objectives and Conflicts

Each stakeholder kept in scope pursues a different objective when an aircraft's trajectory is changed. The airline wants the lowest cost for its flights, which combines fuel, time, and the consequences of a late arrival for connecting passengers. Air traffic control wants separation assured with a workload the sector team can sustain (preferrably minimal), and a traffic picture that stays predictable. The flight crew wants instructions the aircraft can fly, given with enough time to set them up.

Some of these objectives align. Burning less fuel generally lowers the airline's cost and lowers carbon dioxide emissions together. Air traffic control's wish for predictability and the airline's wish to keep its schedule both favor aircraft that fly what was planned.

Others conflict, and the conflict this project studies is the first of them: the trajectory that is cheapest for one aircraft is not always the one that costs the sector least to grant. A request or a resolution that saves fuel for one flight can add instructions, monitoring, and follow-up for the controller. That cost is externalized. It does not appear in the airline's own cost function, and the party that bears it has no means to price it back.

The objectives also operate on different timescales. An airline sets its schedule policy over months, realizes fuel cost per flight, and realizes delay cost in minutes. Air traffic control manages capacity over years through sector design and over seconds through tactical control. A decision made in minutes by one party is judged against objectives the others hold on other clocks.

3.2 Constituent Systems

Stakeholders do not change a trajectory themselves. They act through systems: a controller through the sector's displays and radios, an airline through its operations center and dispatch tools, a crew through the flight deck and the flight management system. And the cost of whatever they decide is actuallized in one place, the aircraft, as fuel burned and time flown. The constituent systems through which the stakeholders act were therefore identified next.

Figure 3 places the NAS at the center and branches outward to the systems that could act on a flight: the aircraft, air traffic control, operators, airports, maintenance suppliers, communication, the regulatory bodies, and passengers. The board is deliberately broad. It follows the system-of-systems view quoted in Section 1, of many heterogeneous systems at several levels, and attempts to list every system that could bear on a flight before any is set aside. Air traffic control divides into the air navigation service providers, whose terminal and en-route facilities control individual flights, and the national command center, whose traffic management units and flow programs shape demand across many. Operators divide into general aviation, airlines and militaries; within the airline, the operations center and its dispatchers sit beside the business, pilots, ticketing and IT functions.

The board decomposes its branches in two different ways, and the difference reflects the problem the project studies. The organizations that decide, air traffic control and the airline, are broken down to the people who make the decision: the individual controller and the dispatcher. The aircraft is broken down to the physical systems that carry it out: the flight management system's guidance and navigation functions, the propulsion components and the fuel system. The decision about a trajectory is made in one part of the system of systems, and its cost is realized in another.

The aircraft is treated alongside the stakeholders as a constituent system. Figure 3 shows the systems first identified within the NAS, with the aircraft decomposed furthest, as its systems and their performance dictate the cost of each flight.

3.3 System of Interest

The System of Interest is drawn from the systems of Figure 3: the systems operated by the four stakeholders kept in Section 3.1.3, and the aircraft. The systems of the boundary actors are not decomposed; they appear only through the context they supply to the decision. The resulting context diagram is Figure 4. It shows the four stakeholders, the aircraft, and the exchanges among them that are relevant to the decision under study.

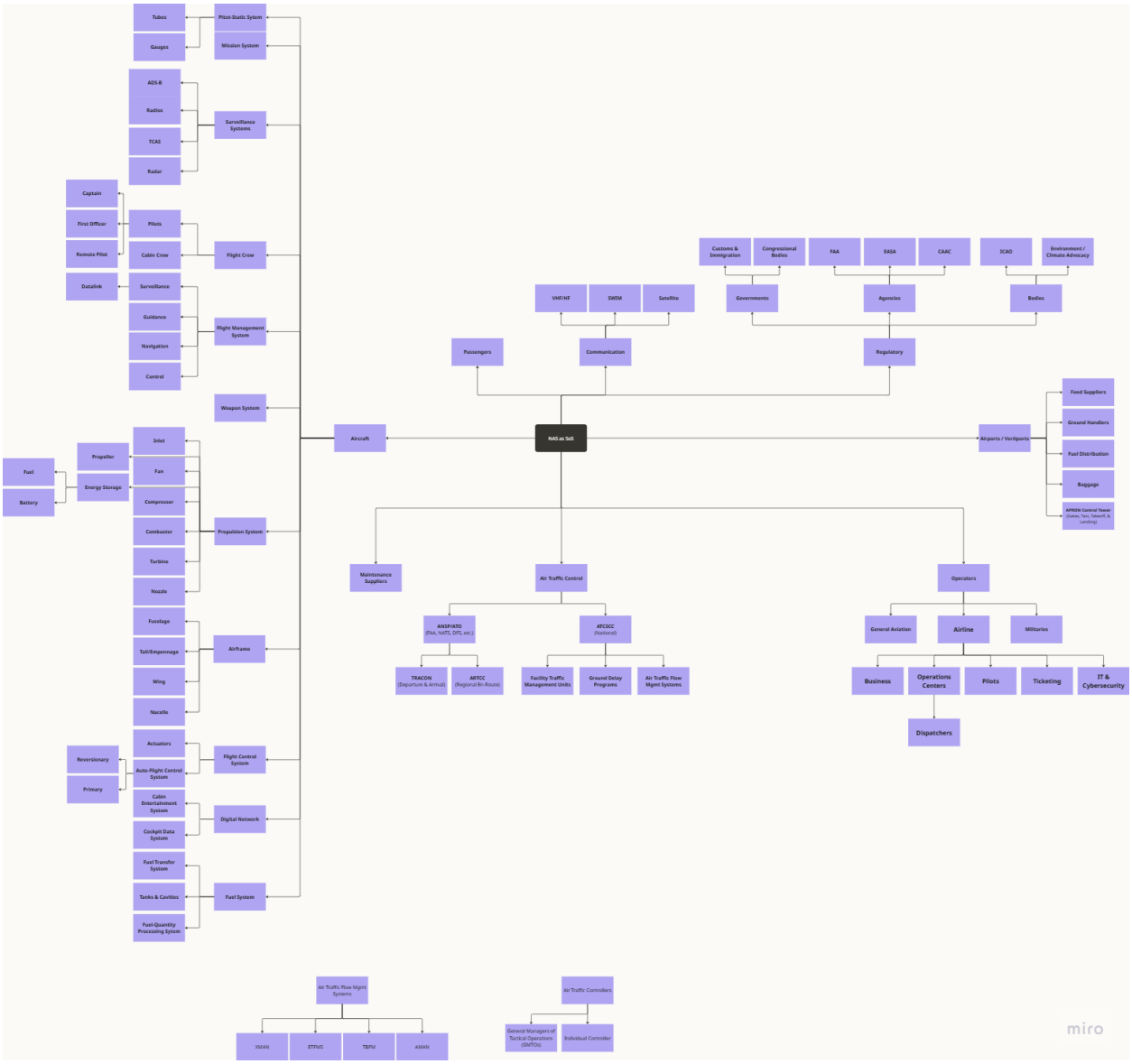


Figure 3: The NAS as a system of systems: candidate constituent systems.

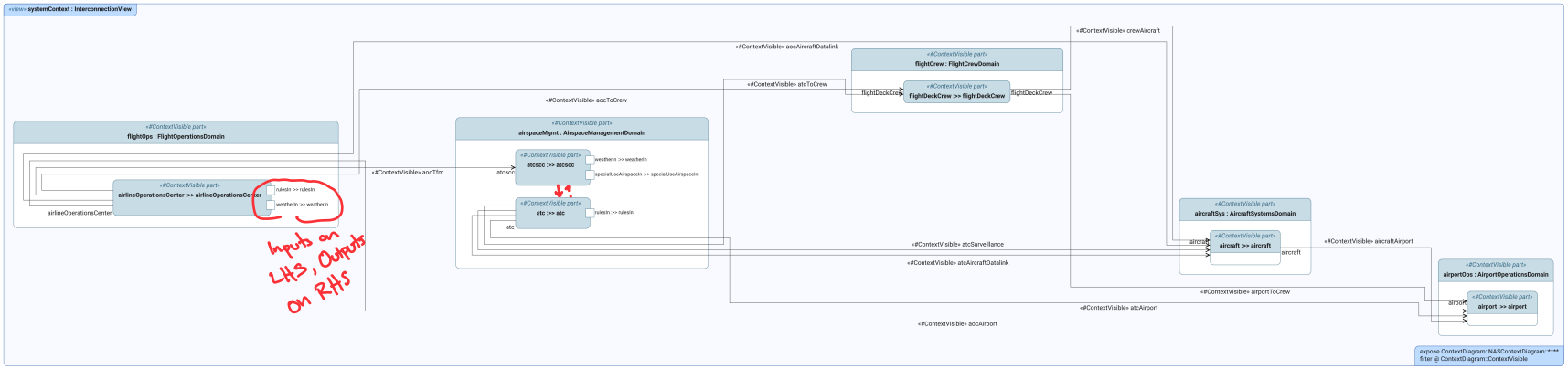


Figure 4: NAS context diagram. View `systemContext`; source `nas_context_diagram.sysml`; rendered 2026-10-10.

AOC
Airport
aircraft
flight crew
NAS Mgmt
all require IBDs.

Output:
cost of NAS

3.4 Operational Concept

The operational concept was first written in full detail as a gate-to-gate thread for one nominal commercial flight, from the pilot's point of view. It runs through eleven phases: the airline's schedule, fleet and crew assignment; the pilot's preparation before duty; reporting and the dispatch release; aircraft preflight and initialization of the flight management system; boarding, loading and fueling; clearance, pushback and taxi; takeoff and initial climb; climb; cruise; arrival planning; and descent through to the gate. Each phase records its initial state, the interfaces in use, and the questions it raised. The full thread is in Appendix A.1.

The model carries an abstraction of that thread. Figure 5 shows six actions in sequence: plan the flight, prepare for departure, depart, fly en route, arrive, and close out the flight. The eleven phases were reduced to six because the study concerns one decision in cruise. The phases before departure and after arrival fix the conditions that decision inherits (the aircraft's weight, its cost index, its schedule position) and are kept at the level needed to show where those conditions come from. Detail is added only where the studied decision is made.

The example conflict sits inside the fly en route action. Two aircraft in cruise are predicted to cross at the same level at the same time. The sector controller must move one of them, and each choice costs fuel, time and instructions that fall on different parties. The nominal thread treats a trajectory change in cruise as one step; the experiment of Section 4 opens that step and asks what the decider knows when making it.

Plan ref: to-do-list.md S5 — Done in the model 2026-10-18: every phase is decomposed, `FlyEnroute` is in Figure 6 with `atcResolveConflict` in the ATC monitoring loop, and the other phases are in Appendix A.1. Still to do: the top-level figure now has seven actions (`Prework` was added first), so the "six actions" paragraph above needs checking; reference Figure 6 in the text and check the conflict paragraph against it.



Figure 5: Nominal flight operations activity diagram, phase level (each phase is decomposed in its own figure). View `nominalOpsActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-18.

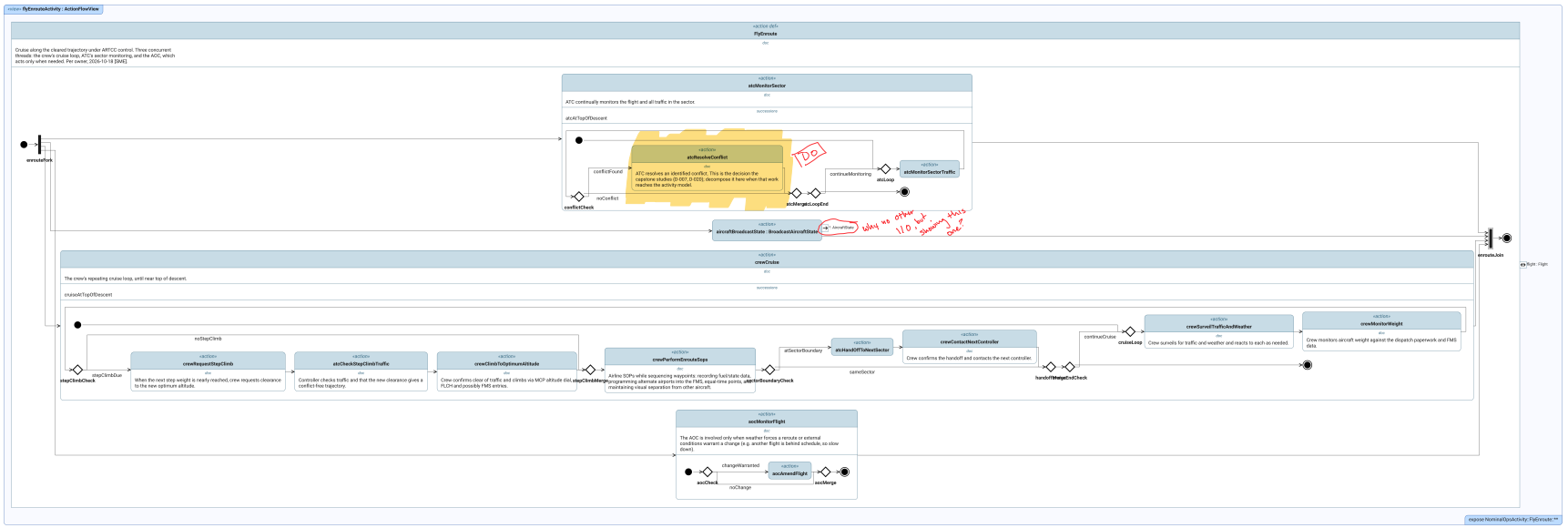


Figure 6: Nominal operations, Fly Enroute: crew cruise loop, ATC sector monitoring and AOC intervention in parallel. View `flyEnrouteActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-18.

Compare w/ GREAT study's output. What's missing? Consider vs. Con Ops. Δ=?

Need to directly connect across other diagrams. At enroute conflict Mgmt, what info has been passed to which systems? Is this something a sub-activity diagram can help highlight? Need to get away from conceptual logic flow & more into actual exchanges, which is IBDs. Does this warrant a single comprehensive IBD of the NAS, or is this going to be straight forward enough to convey across diagrams?

Inputs:

ADS-B position reports

Flight Plans

(MISSING) Perf Data
- source from airlines flight ops

Outputs:

Safe airspace

☐ Need ATC resolveConflict activity diagram!

State diagram for a/c through phases + "in conflict"?

3.5 Architecture: Detailed and Abstracted

Plan ref: to-do-list.md S10 — Present the SysML architecture: package structure, the context diagram (Figure 4), the internal block diagrams below, the interface and item-flow model, and the requirements model with traceability. For the aircraft, the detailed diagram comes first and the abstracted one second (your note: "Place detailed and abstracted"); say what the abstraction drops and why. The airspace management diagram is still a placeholder.

The aircraft is shown at two levels of detail. Figure 7 is the detailed internal block diagram, including propulsion, airframe, landing gear, utilities, and the fuel and power links. Figure 8 is the abstraction used in the rest of the report: the crew, the flight deck controls, the avionics, the flight controls, the fuel system, and the information and command links among them, which are the parts the studied decision passes through. Figure 9 shows the internals of airspace management as modeled so far.

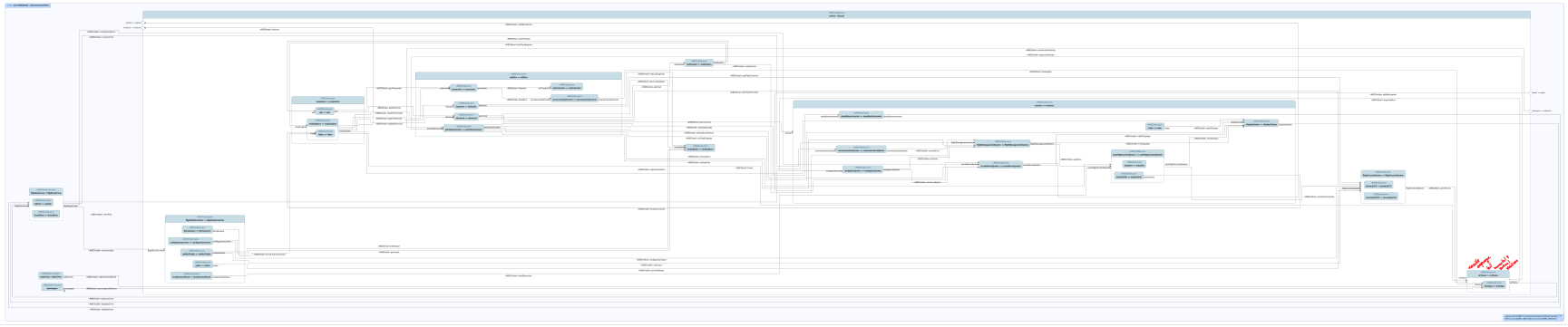


Figure 7: Aircraft internal block diagram, detailed level: adds propulsion, airframe, landing gear, utilities, and the fuel and power links. View `aircraftIbdDetail`; source `nas_ibd_aircraft.sysml`; rendered 2026-10-10.

Output state is important.

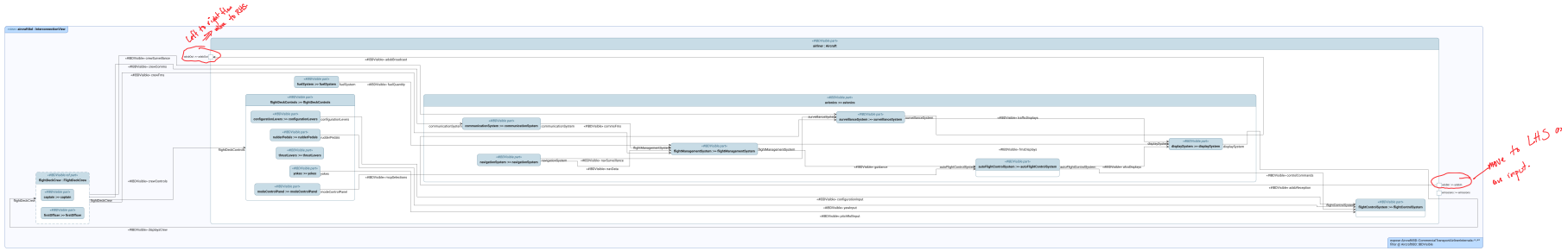


Figure 8: Aircraft internal block diagram, default level of detail. View `aircraftIbd`; source `nas_ibd_aircraft.sysml`; rendered 2026-10-10.

if FCS is included, then engines would have to be too. Landing gear has no effect on cost (except when used in arrival to add drag), so it would be omitted.

II Give justification for every kept item.

- FCS should have internals abstracted, but enacts pilot controls.
- Navigation system determines 4D position & state for sending via comm system.
- Surveillance can be removed.
- Displays can be removed.

I think I'll need to justify each inclusion via an interface that's used in my experiment.

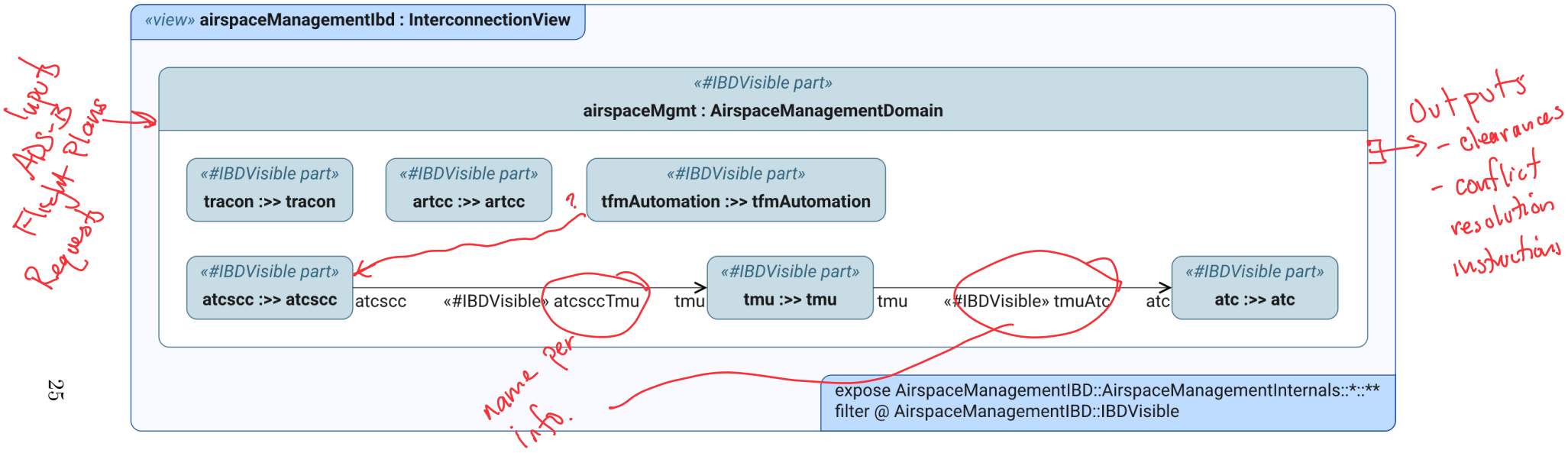


Figure 9: Airspace management internal block diagram. View `airspaceManagementIbd`; source `nas_ibd_airspace_management.sysml`; rendered 2026-10-10.

Need to locate A/C perf info on a diagram or at least be able to point to the port that would have the I/O.

3.6 Proposed Airline-to-ATC Interface

Plan ref: to-do-list.md Track C (D-019) — Name the digital-thread artifact. Describe `AirlinePerformanceShare`: what it carries (gross weight, cost index, connection priority; thrust, drag and fuel-flow model identifiers present but not exercised), who sends it (airline operations center or FMS), who receives it (ATC decision support, per D-012 owned by the system that decides), and which requirements it satisfies (SLR-INF-02) and the goal it serves (SLR-CLR-13). State that security is an assumption. **Owner's note, 2026-10-11 print: "important to do"** — position it against ADS-C EPP and Coppenbarger's Table 2 (the same items, a different receiver); a first draft of that positioning follows.

The items the interface carries are not new. Coppenbarger [15] listed take-off weight, thrust and drag factors, and the speed profile as candidates for exchange between an airline and air traffic control, and measured what airline-supplied weight does for the prediction of a climb. In Europe, the Extended Projected Profile is sent over ADS-C to downlink the trajectory that the flight management system predicts [82]. What has kept the items out of use is ownership, not feasibility: weight and speed intent "are considered competitive parameters by airlines and are not available for use" [79]. A second obstacle is the availability of first-principles data for each aircraft type: thrust, drag and fuel flow.

The proposal here keeps the items and changes the purpose. In the earlier work the data seeks to make the prediction of one aircraft's flight path more accurate; here it feeds the decision support of the party that chooses which aircraft to move, thereby providing for asking "what if" within a mathematical framework. The interface is typed in the model, tagged as part of the improved architecture, and tied to the requirement it satisfies, so the report can point at a model element where the earlier work has a conceptual finding. Confidentiality is stated as an assumption: the data is disclosed only to the deciding party, and no security mechanism is modeled.

The experiment that follows measures what this interface is worth. Its cases differ only in what the decider knows. In cases A and B the decider knows only what air traffic control knows today from broadcast information – the aircraft's type, level, speed and route, from which case B assumes a nominal weight. In cases C and D it also has the items this interface carries. The difference between them is scored against the conflict of Section 3.1.4, at the level of the aircraft, the airline, the sector and the system.

4 Experiment

The experiment tests the airline-to-ATC interface proposed in Section 3.6, resolving around one conflict with and without the information carried by the proposed interface. Each theoretial resolution is scored at different system boundaries, including those of the aircraft, the airline, the sector and the system.

4.1 Scenario

The scenario used to explore the value of the proposed interface is a simple one – four aircraft in one invented en-route sector, shown as one snapshot depicted in Figure 10: a crossing pair at the same level and two aircraft in trail 2,000 ft below.

The crossing pair is at FL380, which places the heavy aircraft is near its service ceiling (Figure 13). The two trailing aircraft are at FL360. The scenario is shown in profile view in Figure 11. The scenario is static, with no time evolution, similar to the studies that informed the scenario ([55], [34]).

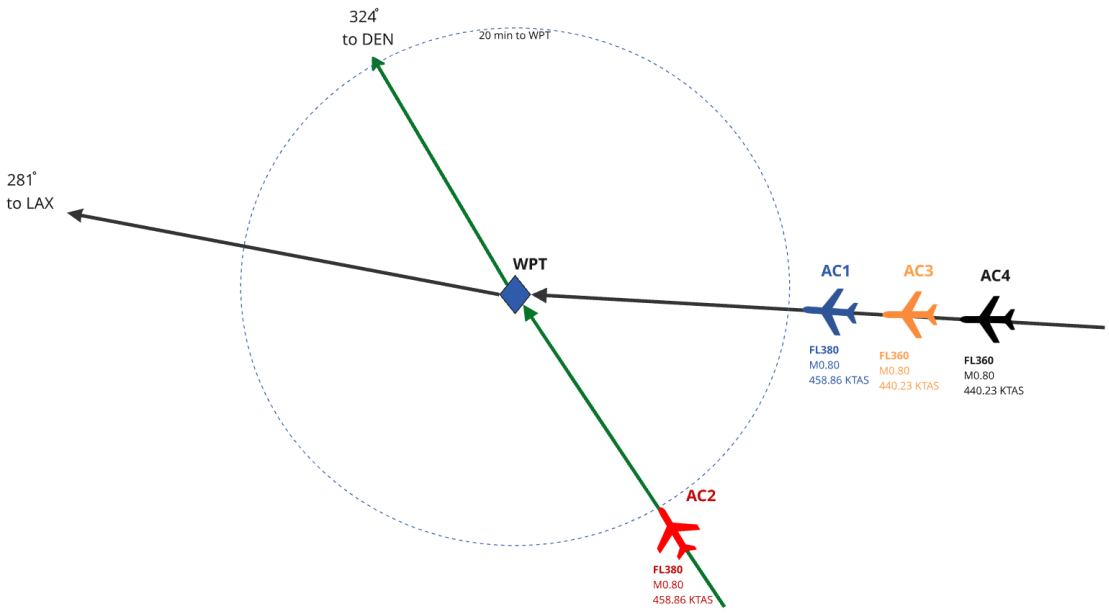


Figure 10: The scenario in plan view. AC1 and AC2 cross at the waypoint at FL380; AC3 and AC4 follow AC1's track at FL360.

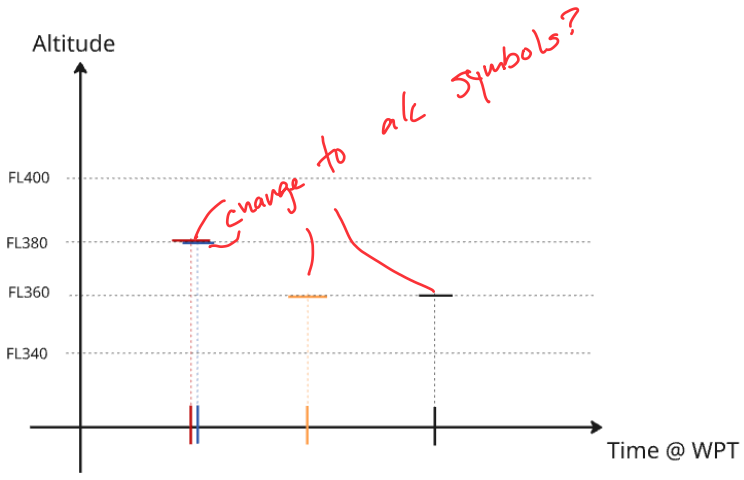


Figure 11: Level of each aircraft against its time at the waypoint. AC1 and AC2 arrive together at FL380.

Plan ref: to-do-list.md S11 (owner's note, 2026-10-11 print, page 11) — "Justify one scenario (time) and expand as possible." Say that one scenario is studied because of the time available in one term, name what a second scenario would add, and list the variations that are run on the one scenario (similar weights, the climb closed, a crossing pair at or below FL350).

4.2 Cases and Resolution Options

Plan ref: to-do-list.md S11 — More realistic comparison? Consider expanding to include a case that mimics EPP, which enables better understanding of the current trajectories and speed envelope (as evidenced by ETA windows). What else can be derived from EPP info in a split-second assessment? Also, should we throw "typical weight errors" into the tool and see their impact on the system's ability to make good decisions? How might that factor into the model?

Four cases are considered of varying levels of apriori information, and are summarized in Table 4. The decider's knowledge of the aircraft's perfomance and the relative expense of airline costs determines which resolution options it can offer.

Table 4: Conflict resolution options available to a controller with different levels of information about the aircraft.

Case	Decider	Information	Climb rule applies?
A	Controller, by rule	Type, level, speed, route	Yes
B	Controller, with decision aid	Nominal weight per type	Yes
C	Controller, with decision aid	Shared weight and cost index	No
D	Controller, with decision aid	Shared weight, cost index and delay margin	No

Case A is the baseline, which is the current NAS practice. It has only the information that is currently available to the controller through current interfaces, which limits the options it can offer. [19] observed that controllers prefer to descend aircraft rather than climb them, and that they will not offer a climb if the aircraft is above FL350. The baseline case applies this rule, which limits the options it can offer. [73] found that a single descent advisory was the most common resolution option.

Case B is an automated decision aid that has access to the nominal weight of each aircraft type, but not the actual weight or cost index. As such, the system does not have the information needed to guarantee that a climb is feasible. If an unachievable climb is proposed, the flight crew would have the obligation to reject it, leading to unacceptable delays in conflict resolution. As such, the system must also apply the climb rule in Case B.

Case C is an automated decision aid that has access to the actual weight and cost index of each aircraft. As such, it is not bound by the same limitations as Case B. It differs from Case D through the ability to account for the cost of delaying the time of arrival of the aircraft. The cost calculation os Case D incorporate a measure of the cost of delaying a flight with connections past its threshold, which is not available to Case C.

Plan ref: to-do-list.md S11 — Resolution options. State which aircraft descends and the tie-breaker (undecided), and whether the aircraft is cleared back up. For each option give the instructions it costs now, the follow-up it creates, and what the decider must know to use it. Cite [34] for the instruction accounting.

The conflict depicted in the scenario can be resolved by several options, which are listed in Table 5 and are available to all four cases, with the exception of options 3 and 4, which are not available for cases A and B for reasons described previously.

Table 5: Resolution options. A dash means no instruction. Full version with instruction counts in Appendix A.5.

Option	AC1	AC2	AC3	AC4
O1	Descend 2,000 ft	–	–	–
O2	–	Descend 2,000 ft	–	–
O3	Climb 2,000 ft	–	–	–
O4	–	Climb 2,000 ft	–	–
O5	Turn behind AC2	–	–	–
O6	–	Turn behind AC1	–	–
O7	Reduce speed	–	–	–
O8	Descend 4,000 ft	–	–	–
O9	Descend 2,000 ft	–	Slow or turn	–

Options 1 and 2 are descents of the heavy and light aircraft, respectively. A descent is the most common resolution option observed in practice [73]. It is the safest option, as it is robust to heavy aircraft, which may be operating near their service ceiling with little room to maneuver. Cases A and B will be unable to discern a difference in cost between the two options due to limited knowledge.

Options 3 and 4 are climbs of the heavy and light aircraft, respectively, which are not available to cases A and B due to the climb rule. If A and B were able to consider climbs they would be similarly ignorant to the relative benefit and safety associated with the lighter aircraft climbing.

In Options 5 and 6, a lateral modification is assigned by the controller to achieve separation without the need for changes to flight level. Option 7 considers a speed reduction of the heavy aircraft, which is a less common resolution option, but which is practical and straight forward way to achieve separation if there is sufficient separation on either side of the pair. Aircraft 3 and 4 complicate its application to this scenario.

Option 8 is a 4,000 ft descent of the heavy aircraft, which a controller may deem necessary to simplify the separation with aircraft 3 and 4, which are in trail 2,000 ft below aircraft 1. Option 9 is a descent of the heavy aircraft and a speed reduction or turn of the light aircraft, which is a less common resolution option.

While not comprehensive, this set of options is representative of controller practices. [34] found that controllers relied on five lateral vectoring heuristics and eight vertical level-change heuristics to resolve potential conflicts between aircraft. [73] found that level-offs constituted the most frequent maneuver category, which are only applicable to aircraft ascending or descending and which represented 44%. Turns were the next most frequent, at 32%, followed by descents at 18%, and climbs at 5%. Turns occurred much less frequently than vertical trajectory changes, accounting for 32% compared with 68%. These percentages reflect actions taken across all types of conflicts, rather than the cruise scenario alone.

Plan ref: to-do-list.md S11 — To Specify degree of turn, distance offset, and re-vectoring back on path, whether direct-to or otherwise. This will manifest in different costs.

4.3 Scoring

Each option is scored along several lines, to highlight the ways different actions lead to different costs and how those costs aggregate at varying system of interest boundaries, including:

- a single aircraft,

- all aircraft associated with a single airline,
- all aircraft in the sector,
- the airspace as a whole, including the cost of controller actions, airline schedule impacts, and environmental impacts.

Plan ref: to-do-list.md S11 — Review items. Still need to be checked against `simulation/experiment_scoring.py`. Also, need to finalize what costs will be included where, based on the models. Current content is a draft.

Individual aircraft costs are limited to time and fuel, and are equated through the use of a standard industry conversion with the cost index (CI). For aircraft i the cost of an option is its extra fuel plus its cost index times its extra time, in kilograms of fuel, the form used in [69]:

$$c_i = \Delta m_{f,i} + CI_i \, \Delta t_i, \tag{1}$$

where the extra fuel $\Delta m_{f,i}$ and the extra time Δt_i are the option minus the baseline of flying on undisturbed, and CI_i is the cost index of aircraft i in kilograms of fuel per minute.

An airline's cost is the sum over its aircraft plus a step cost when a flight misses a connection, plus CO2 converted to dollars.

The cost experienced by the sector include all individual aircraft costs as well as clearances issued, effort to devise them, and residual risk. The latter three costs are associated with the sector

Total airspace cost is that of the system of interest, and totals all costs experienced by airspace users and by the stakeholders of the system. As such, it includes the environmental cost associated with each action. This would also consider the balance of costs across airlines, if the scenario facilitated resolutions that lead to inequitable costs.

Plan ref: to-do-list.md S11 — Emissions. State the conversion from fuel to CO2 and its source ([48]). State how CO2 becomes dollars at the airline level and why that route was chosen (credit price, passenger value, or both; undecided), and why the system level keeps CO2 in kg.

4.3.1 Case Considerations

Case A applies its rule, one descent. Each automated case chooses, among the options it is allowed to offer, the option whose estimated total of Eq. 1 over all aircraft is lowest. Case B uses the type's nominal mass for every aircraft and values time at 1 kg/min. Cases C and D use each aircraft's actual mass and cost index. Case D also rejects an option that delays a flight with connections past its threshold. Whatever a case chooses is then scored at the actual masses, and that score is what the tables report.

Schedule reliability enters as a step cost, as described by [72], which occurs when a flight with connections is delayed past a given threshold. At the time threshold associated with missed connections, the flight incurs a cost K on top of Eq. 1. No sourced value of K is presently held, so the tables report whether the threshold is crossed, not a dollar figure. This may change in a future draft.

Carbon dioxide can be derived from the chemical formula of Jet-A fuel, as 3.16 kg per kilogram of fuel [48]. Instructions issued by controllers are counted for those required to resolve the initial conflict as well as those to clear the aircraft back to its original path, but are not presently converted to a USD cost.

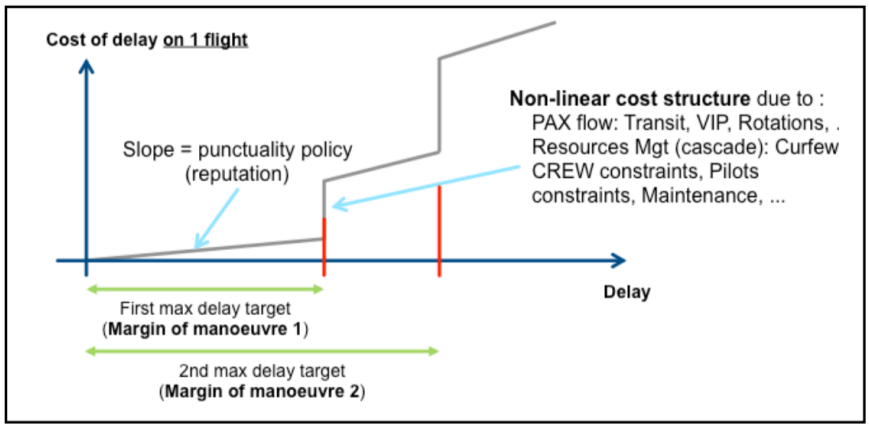


Figure 1: Typical cost-delay model profile per flight

Figure 12: Cost of delay for one flight, with steps where a margin is exceeded. Reproduced from [72], Figure 1.

4.4 Aircraft Model and Its Simplifications

Calculating a fuel and time cost involves defining a truth model to enable comparison of the resultant airspace performance observed as a result of resolving conflicts using the various optoins. The OpenAP (open aircraft performance model, TU Delft) is used to define that entitled information and supplies fuel flow, thrust and drag for the scenario aircraft, which are combined in the equations of motion to determine fuel and time deltas.

Each aircraft is a point mass in steady flight at a constant Mach number. Drag, maximum thrust and fuel flow are taken from OpenAP [88], which uses a two-term drag polar and, above 30,000 ft, the two-shaft turbofan thrust model of Bartel and Young [3].

Plan ref: to-do-list.md S11, S12 — OpenAP Review Read Sections 5 and 7 of the OpenAP paper (pp. 6 to 17) before submission. What the paper says about its own accuracy is in `evidence/literature-notes/summaries/4 - sun2020openap.md` and belongs in the discussion of the truth model's uncertainty (Section 5.2).

A descent is flown at 1,500 ft/min. A climb is flown at maximum thrust, at the rate that thrust allows and no more than 1,500 ft/min. A level is available to an aircraft only if it would still have 150 ft/min of climb left there. Cases A and B do not know the mass and so cannot make that test; they offer a climb only at or below FL350, which the scenario does not explore. Cases C and D make it at the aircraft's actual mass. Figure 13 depicts a typical flight envelope, drawn assuming lift equals weight and using the OpenAP models.

The low-speed boundary is 1.3 times the stall speed at an assumed clean maximum lift coefficient, and its high-speed boundary is the more restrictive of the maximum operating Mach number and the maximum operating speed (VMO). The aircraft's present level and Mach number are marked, and each altitude and speed instruction the scenario offers it is marked as inside or outside the region with 150 ft/min of climb left. Speed changs of +/- 0.04 Mach are presented, as these are typical simplifications employed by European controllers. The weight chosen for the figure shows the limitations associated with the speed simplifications, both in their inability to use the full speed change authority available to the aircraft and to forsee rejections that pilots will invevitably convey upon clearance issuance.

The model makes the following simplifications.

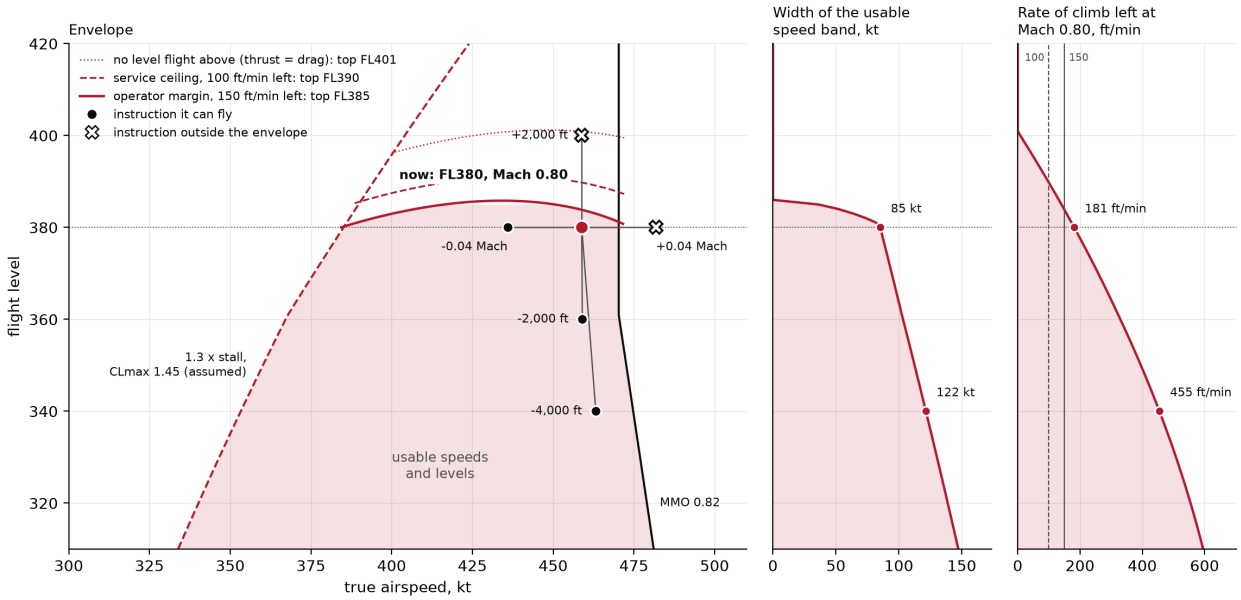


Figure 13: The heavy B737-9 (71.0 t) has little room left to change speed or level at FL380. Left: cruise envelope, with three lines of constant rate of climb left: 0 ft/min (thrust equals drag), 100 ft/min (the service ceiling, [22]) and 150 ft/min (the operator margin the scenario uses). Right: width of the usable speed band and rate of climb left at Mach 0.80, by flight level.

1. **Standard atmosphere, no wind, lift equal to weight.** Time differences come only from true airspeed and added track distance.

2. **No compressibility drag.** The drag used here has no Mach-number term, though its supported by OpenAP and has a non-trivial impact at Mach numbers above 0.75.

3. **The drag polar is the 737 MAX 8's.**

4. **The fuel curve is the package default.** OpenAP has no fuel model fitted to the 737 MAX 9 or MAX 8; it uses its default coefficients scaled by the LEAP-1B take-off fuel flow.

5. **The climb margins.** The service ceiling is the level with 100 ft/min of climb left [22]. The 150 ft/min the experiment requires is commonly observed in practice.

6. **One climb rate per climb.** The rate is taken at the mid-altitude and held; the climb is not integrated. A climb is not checked against the time left before the crossing.

7. **Mass is constant.** Fuel burned during the manoeuvre does not reduce the mass.

8. **The low-speed limit rests on an assumed maximum lift coefficient** ($C_{L\,\max} = 1.45$) and is a stall margin, not a buffet boundary.

5 Results and Discussion (Preliminary)

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Results and Discussion (preliminary). For Interim Report #2 this section needs: (1) the architecture as it stands (context diagram, aircraft IBD, the three requirements tagged `improved` as the measured gap) — the figures are now in Section 3; (2) the two hand checks from `experiment-definition.md` §9 written up as preliminary results; (3) the first OpenAP numbers — Figure 14 and Table 6 are in place, the discussion is not; (4) one paragraph on what is not yet done.

5.1 Model Results

Plan ref: to-do-list.md S10, S13 (D-028) — Model results. With the architecture now presented in Section 3, report its measures here, against the measures named in Section 1.5: for example, how many stakeholder requirements trace to a kept objective, how many system requirements are allocated to a constituent system, and which requirements are tagged `improved` because today's NAS does not meet them. Without this, the systems engineering work is described but never assessed.

5.2 Experiment Results

Table 6: Outcome of each case at four levels, for the base scenario scored at the actual masses. Each cost is a difference from flying on undisturbed. Preliminary: the scenario's numbers are not all confirmed.

Case	Option	Largest cost to any one aircraft (kg fuel-equiv.)	Airline totals (USD)	Sector (calls / effort / risk)	System fuel, CO2 (kg)
A	O8	32 (AC1)	X 28, Y 0, Z 0	1+1 / 1 / low	32, 101
B	O9	11 (AC1)	X 15, Y 0, Z 0	2+2 / 3 / medium	16, 51
C	O4	12 (AC2)	X 0, Y 11, Z 0	1+1 / 3 / low	12, 39
D	O4	12 (AC2)	X 0, Y 11, Z 0	1+1 / 3 / low	12, 39

Plan ref: to-do-list.md S12 — Which option each case chose, and why. Walk Table 6 row by row. For case A, name the rule that produced the choice. For B, C and D, name the cost that decided it.

Plan ref: to-do-list.md S12 — Did the climb rule matter? With the crossing pair above FL350, cases A and B could not offer a climb and cases C and D could. Say whether case C used one, for which aircraft, and what it saved. If C did not climb either, say so: that is a result too.

Plan ref: to-do-list.md S12 — B minus A, and C minus B. The first is what automation is worth; the second is what the information exchange is worth. Compare the size of each with the prior figure of tens of gallons per flight ([19], p. 12) and say whether the difference is larger than the truth model's uncertainty.

Plan ref: to-do-list.md S9, S12 — Does the best option change with the level? Read across each row of Table 6. Name any option that is better for one aircraft or one airline and worse for the sector or the system, and who pays (the largest cost to any one aircraft). This is the sub-system versus system discussion.

Plan ref: to-do-list.md S12 — Airline X's two flights. Discuss the option that touches both of one airline's aircraft: what the airline would choose between its late flight and its early one, and that the controller cannot see that preference. Connect to the dispatcher priorities (schedule integrity and connections rated at or above fuel; source of that table still to confirm).

Plan ref: to-do-list.md S12 — Workload is three things. Discuss clearances issued, effort to devise, and residual risk separately. Say where a "more workload" option on one of them is "less" on another.

Plan ref: to-do-list.md S12 — When would the controller say no? (owner's note, second hand review, 5.2). Discuss whether refusals are more likely in morning rushes at hub airports or near holidays, using the sector-loading rule in [19] (p. 10). A hypothesis; needs a source.

Plan ref: to-do-list.md S12 — Is schedule the right priority? (owner's note, second hand review, 4.11). Contrast a carbon price, which makes fuel weigh more against time, with controllers' usual practice of giving schedule the priority.

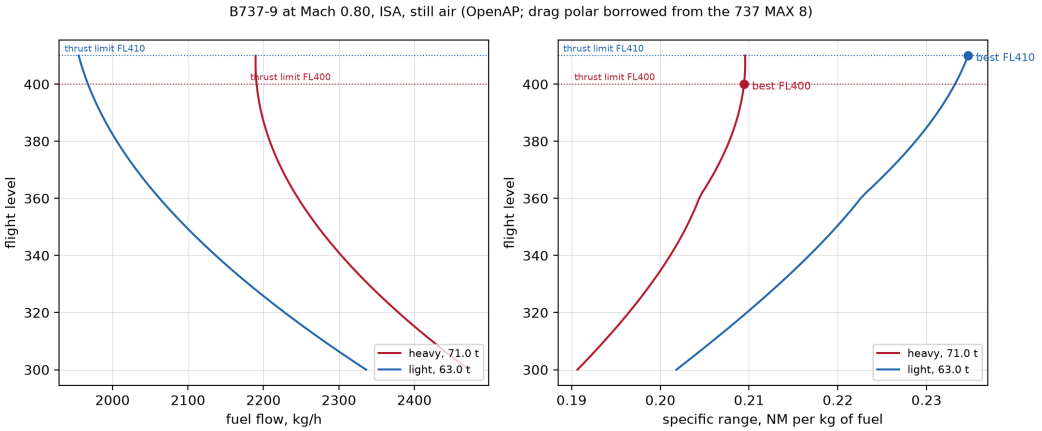


Figure 14: Fuel flow and specific range against altitude at Mach 0.80 for the heavy and light B737-9 (OpenAP).

Plan ref: to-do-list.md Track A — Figure slot: sequence diagram of case C's conflict resolution (controller, ATC decision support, airline operations center, FMS, crew), rendered from the model once drafted (turn 3).

Plan ref: to-do-list.md S12 — The case where nothing changes. Report the variation with similar weights and the climb closed, where case C picks the same option as case B, to show the comparison is not rigged.

Each column of Table 6 belongs to a part of the context diagram (Figure 4). The aircraft column belongs to the aircraft and its flight crew. The airline column belongs to the airline operations center within flight operations. The sector column belongs to air traffic control within airspace management. The system column belongs to no single part. It is a property of the NAS as a whole, which is the reason no constituent system computes it today and the reason the model is needed to state it.

The exchange that cases C and D add is one interface in the model: the airline operations center's performance share to air traffic control's decision support, carrying gross weight, cost index and the flight's delay margin. It satisfies system requirement SLR-INF-02, which is tagged as part of the improved architecture because today's NAS does not meet it. The interface is not drawn on Figure 4, which shows the baseline.

5.3 Emissions and Who Pays

Plan ref: to-do-list.md S9, S12 — Airline level: put CO2 into dollars. Take the fuel difference between cases from Table 6, convert to kg of CO2, then to dollars. State the route used and its source. Candidates: (1) the price of a carbon credit or allowance, if a scheme applies to the flight; (2) passenger value, meaning revenue gained or lost because some passengers choose the flight listed with lower emissions. Make the point that fuel is already counted once as fuel cost and that pricing its CO2 counts the same kilograms again, on purpose.

Plan ref: to-do-list.md S9, S12 — Passenger value. Discuss how passengers see a flight's emissions when booking, and whether that reaches the airline as demand. The author's own practice of choosing lower-emission listings is the motivation, not the evidence: cite a study. Note the gap between one decision in one sector and the label a passenger saw at booking; the link is the airline's average over many flights.

Plan ref: to-do-list.md S12 — Scale it honestly. One conflict in one sector is a small number. If a per-year or fleet-wide figure is offered, state the multiplier and its source (for example, how often level crossing conflicts occur), and label it illustrative.

Plan ref: to-do-list.md S9 — Who pays. For the option the system prefers, name the aircraft or airline that is worse off and by how much. This is the equity discussion.

6 Impact/Financial Benefits

Plan ref: to-do-list.md S9, S12 (S16 bullet added this session) — Quantify the impact of the reference architecture and its experiment. Translate the conflict analysis and the experiment's results into concrete operational terms wherever possible (e.g., delay-minutes, fuel burn, CO2, instructions issued) and into US-dollar terms where a credible conversion exists (e.g., published cost-per-delay-minute figures from the literature review, Section 1.2). Also discuss the ethical context and any unintended consequences of the recommendations (equity-minded engineering: whose costs get externalized, and onto whom – this connects directly to the objectives whose costs are externalized onto another stakeholder, Section 3.1.4, and to Section 5.3).

Scope flag: the 503 template expects a quantified financial-benefit statement (the template's sample statements are dollar-savings claims from industry DMAIC-style projects). This capstone is an architecture/SoS project, not a cost-reduction project with a "before" baseline in dollars, so a literal dollar-savings figure may not be available or meaningful. Recommend framing impact primarily in operational/systems terms (delay, fuel, emissions, predictability, tradeoff-surface shape) with a dollar estimate offered as an illustrative, clearly-caveated secondary figure rather than the headline metric.

7 Conclusions and Recommendations

Plan ref: to-do-list.md S15, S16 — Briefly restate the problem, approach, and findings, including potential benefits in relative terms if absolute figures are sensitive or incomplete (e.g., "reduces network-level delay propagation by an illustrative X% under the demonstrated scenario" rather than an unsupported absolute claim). Cover: known architecture omissions, limitations of the selected decomposition (S6), limitations of the experiment (Sections 2.2 and 4.4), and opportunities for future expansion (e.g., additional off-nominal scenarios, unexplored S9 conflicts, deeper crew/maintenance-routing integration).

Plan ref: to-do-list.md S14, S15 — Recommendations. Restate the main findings from Section 5 and provide recommendations for moving forward. Include the alternatives considered (alternative stakeholder-weighting schemes from S11) and the criteria used to select the recommended approach. Suggest additional areas of study (candidate: architecture extensions, additional off-nominal scenarios from S15, unexplored myopic-optimization conflicts from S9's catalog).

References

[1] Airbus Customer Services. *Flight Operations Briefing Notes: Approach Techniques: Aircraft Energy Management during Approach*. Tech. rep. FLT_OPS – APPR – SEQ03 – REV02. Airbus, Oct. 2005. URL: `https://skybrary.aero/sites/default/files/bookshelf/166.pdf`.

[2] Magnus Axholt and Stephen Peterson. “Modelling Traffic Scenarios for Realistic Air Traffic Control Environment Testing”. MA thesis. Linköping University, 2004.

[3] Matthias Bartel and Trevor M. Young. “Simplified Thrust and Fuel Consumption Models for Modern Two-Shaft Turbofan Engines”. In: *Journal of Aircraft* 45.4 (2008). Not held and not read. Authors, title, journal, volume, issue, pages and DOI confirmed 2026-10-11 against the Crossref record for the DOI (DOI supplied by the owner)., pp. 1450–1456. DOI: 10.2514/1.35589.

[4] Jason E. Bartolomei et al. “Engineering Systems Multiple-Domain Matrix: An Organizing Framework for Modeling Large-Scale Complex Systems”. In: *Systems Engineering* 15.1 (2012), pp. 41–61. DOI: 10.1002/sys.20193.

[5] Katherine A. Berry and John J. Pace. “Examining the Actors and Functions of an Airline Operations Center”. In: *Proceedings of the Human Factors and Ergonomics Society 55th Annual Meeting*. Vol. 55. 1. 2011, pp. 1412–1416. DOI: 10.1177/1071181311551294.

[6] Karl D. Bilimoria, Mike M. Paglione, and Hilda Q. Lee. “Performance Analysis of a Conflict Probe Utilizing Only State Vector Information”. In: *24th International Congress of the Aeronautical Sciences (ICAS)*. Yokohama, Japan, 2004. URL: https://www.icas.org/icas_archive/ICAS2004/PAPERS/216.PDF.

[7] Katie Bolczak et al. *FAA Data Standards Initiative: Systems Engineering Base for Air Traffic Modernization*. Tech. rep. The MITRE Corporation, 2001.

[8] Soufiane Bouarfa, Jasper Müller, and Henk A. P. Blom. “Evaluation of a Multi-Agent System Approach to Airline Disruption Management”. In: *Journal of Air Transport Management* 71 (2018), pp. 108–118. DOI: 10.1016/j.jairtraman.2018.05.009.

[9] Chris Brady. *Rules of Thumb*. The Boeing 737 Technical Site (web page). 2026. URL: http://www.b737.org.uk/rulesofthumb.htm.

[10] Ian Britchford. “737 MAX 8 & 9 fuel burn and operating performance”. In: *Aircraft Commerce* 137 (Aug. 2021), pp. 18–26. URL: https://www.aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs/General%20Articles/2021/137_FLTOPS.pdf.

[11] António Jesus Monteiro de Castro. “A Distributed Approach to Integrated and Dynamic Disruption Management in Airline Operations Control”. PhD thesis. Porto, Portugal: Faculdade de Engenharia, Universidade do Porto (FEUP), 2013.

[12] Mohammad Chami et al. “Towards Solving MBSE Adoption Challenges: The D3 MBSE Adoption Toolbox”. In: *28th Annual INCOSE International Symposium*. Washington, DC, 2018.

[13] Gano Chatterji. *Introduction to Air Traffic Management*. Video and transcript, Aviation Data Science Seminar (virtual). NASA Technical Reports Server (NTRS), Document ID 20205001251. NASA Ames Research Center; Universities Space Research Association, Apr. 2020. URL: https://ntrs.nasa.gov/citations/20205001251 (visited on 09/20/2026).

[14] Michael D. D. Clarke. “Irregular Airline Operations: A Review of the State-of-the-Practice in Airline Operations Control Centers”. In: *Journal of Air Transport Management* 4 (1998), pp. 67–76.

[15] Richard A. Coppenbarger. “En Route Climb Trajectory Prediction Enhancement Using Airline Flight-Planning Information”. In: *AIAA Guidance, Navigation, and Control Conference and Exhibit*. AIAA-99-4147. 1999, pp. 1077–1087. DOI: 10.2514/6.1999-4147.

[16] Daniel A. DeLaurentis. "Understanding Transportation as a System-of-Systems Design Problem". In: *43rd AIAA Aerospace Sciences Meeting and Exhibit*. Reno, Nevada, 2005. DOI: 10.2514/6.2005-123.

[17] Daniel A. Delaurentis and Donald N. Fry. "Understanding the Implications for Airports of Distributed Air Transportation Using a System-of-Systems Approach". In: *Transportation Planning and Technology* 31.1 (2008), pp. 69–92. DOI: 10.1080/03081060701835738.

[18] Ahmed E. E. Eltoukhy, Felix T. S. Chan, and S. H. Chung. "Airline Schedule Planning: A Review and Future Directions". In: *Industrial Management & Data Systems* 117.6 (2017), pp. 1201–1243. DOI: 10.1108/IMDS-09-2016-0358.

[19] Engility Corporation. *Traffic Aware Strategic Aircrew Requests (TASAR): Annualized TASAR Benefits for Alaska Airlines Operations*. Contractor report NNL12AA06C, deliverable 41A. Corporate author, as the document states: "prepared by Engility Corporation ... under Contract No. NNL12AA06C with NASA Langley Research Center" (Preface, p. 2); NASA Technical Monitor: David Wing (title page). No personal author is named. A web search summary of the NASA record for a later published version (NASA/CR-2015-218787, NTRS 20150017048) names Jeffrey Henderson; not confirmed and not used here. Engility Corporation, Billerica, MA, for NASA Langley Research Center, Sept. 2014. URL: https://ntrs.nasa.gov/api/citations/20140012787/downloads/20140012787.pdf.

[20] EUROCONTROL. *EUROCONTROL Specification for Airport Collaborative Decision Making (A-CDM)*. Tech. rep. Brussels, Belgium: EUROCONTROL, 2025.

[21] EUROCONTROL. *Aircraft Performance Database: BOEING 737 MAX 9 (B39M)*. Web page. 2026. URL: https://learningzone.eurocontrol.int/ilp/customs/ATCPFDB/details.aspx?ICAO=B39M.

[22] Federal Aviation Administration. *Aircraft Operations at Altitudes Above 25,000 Feet Mean Sea Level or Mach Numbers Greater Than .75*. Advisory Circular AC 61-107B, Change 1. Washington, DC: Federal Aviation Administration, U.S. Department of Transportation, Sept. 2015. URL: https://www.faa.gov/regulations_policies/advisory_circulars.

[23] Federal Aviation Administration. *Instrument Procedures Handbook, Chapter 3: Arrivals*. Tech. rep. FAA-H-8083-16B. U.S. Department of Transportation, Federal Aviation Administration, 2017. URL: https://www.faa.gov/sites/faa.gov/files/regulations_policies/handbooks_manuals/aviation/instrument_procedures_handbook/FAA-H-8083-16B_Chapter_3.pdf.

[24] Federal Aviation Administration. *Flight Information Exchange Model (FIXM) US Extension v4.4.0: Logical Model Diagrams*. Tech. rep. Federal Aviation Administration / FIXM Program, 2024.

[25] Federal Aviation Administration. *Aeronautical Information Manual: Official Guide to Basic Flight Information and ATC Procedures*. Aeronautical Information Manual. Washington, DC: Federal Aviation Administration, U.S. Department of Transportation, Feb. 2025. URL: https://www.faa.gov/air_traffic/publications.

[26] Federal Aviation Administration. *Air Traffic Control*. Order JO 7110.65BB. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[27] Federal Aviation Administration. *FAA Services Hierarchy*. One-page slide (PDF), FAA NAS Enterprise Architecture. Year is the PDF creation date (2025-02-24, PowerPoint export; PDF author field "ANG-B2"); no version, title-page date or byline appears on the slide. Unverified: what ANG-B2 stands for, and whether this is the current issue of the hierarchy on the FAA NAS EA site (faa.gov/about/office_org/headquarters_offices/ang/r2r) – check before citing as current. 2025.

[28] Federal Aviation Administration. *Facility Operation and Administration*. Order JO 7210.3EE. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[29] Federal Aviation Administration. *Flight Services*. Order JO 7110.10EE. Washington, DC: Federal Aviation Administration, Air Traffic Organization Policy, Feb. 2025.

[30] Federal Aviation Administration. *NAS Enterprise Architecture: Infrastructure Roadmaps*. Tech. rep. Federal Aviation Administration, 2025.

[31] Federal Aviation Administration. *Airport Compliance Manual, Chapter 12: Review of Aeronautical Lease Agreements*. Order 5190.6C. URL not recorded when the chapter was downloaded on 2026-09-28; add it from the FAA Order 5190.6C page before citing. Washington, DC: Federal Aviation Administration, Office of Airports Compliance and Management Analysis, Feb. 2026.

[32] Federal Aviation Administration. *Airport Compliance Manual, Chapter 9: Unjust Discrimination between Aeronautical Users*. Order 5190.6C. Washington, DC: Federal Aviation Administration, Office of Airports Compliance and Management Analysis, Feb. 2026. URL: https://www.faa.gov/airports/resources/publications/orders/compliance_5190_6/order-5190-6c-compliance-chapter-09 (visited on 09/28/2026).

[33] Michael Finke et al. *D2.1: Current TBO Concepts and Derivation of the Green Air Traffic Management Concepts*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2020.

[34] Selina Fothergill and Andrew Neal. "Conflict-Resolution Heuristics for En Route Air Traffic Management". In: *Proceedings of the Human Factors and Ergonomics Society 57th Annual Meeting*. 2013, pp. 71–75. DOI: 10.1177/1541931213571018.

[35] GE Aerospace. *GE Aerospace Joins MBSE Ecosystem in Michigan Aerospace*. University of Michigan College of Engineering, Aerospace Engineering news. May 2023. URL: https://aero.engin.umich.edu/2023/05/11/ge-aerospace-joins-mbse-ecosystem-in-michigan-aerospace/ (visited on 09/14/2026).

[36] L. K. Hassan, B. F. Santos, and J. Vink. "Airline Disruption Management: A Literature Review and Practical Challenges". In: *Computers & Operations Research* 127 (2021), p. 105137. DOI: 10.1016/j.cor.2020.105137.

[37] Terry R. Hill et al. "NASA's Use of MBSE and SysML Modeling to Architect the Future of Human Exploration". In: *34th Annual INCOSE International Symposium*. Dublin, Ireland, 2024.

[38] Eric C. Honour. "Systems Engineering Return on Investment". In: *INCOSE International Symposium*. Vol. 20. 1. Chicago, IL, 2010, pp. 1422–1439. DOI: 10.1002/j.2334-5837.2010.tb01150.x.

[39] Eric C. Honour. "Sizing Systems Engineering Activities to Optimize Return on Investment". In: *INCOSE International Symposium*. Vol. 21. 1. Denver, CO, 2011, pp. 1233–1245. DOI: 10.1002/j.2334-5837.2011.tb01281.x.

[40] Eric C. Honour. "Systems Engineering Return on Investment". PhD thesis. Adelaide, Australia: Defence, Systems Institute, School of Electrical, and Information Engineering, University of South Australia, 2013. URL: https://segoldmine.ppi-int.com/node/72248 (visited on 09/19/2026).

[41] Niamat Ullah Ibne Hossain et al. "Modeling and Analysis of Unmanned Aerial Vehicle System Leveraging Systems Modeling Language (SysML)". In: *Systems* 10.6 (2022), p. 264. DOI: 10.3390/systems10060264.

[42] Haoliang Hu et al. *D2.2: Report about Modelling of Operational and System Architecture*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2021.

[43] Haoliang Hu et al. *D5.1: Report on ATM Avionic System Architecture Development*. Tech. rep. GreAT (Greener Air Traffic Operations) Project, EU Horizon 2020 / China RIA, Grant Agreement 875154, 2021.

[44] Yuzhen Hu et al. "Review of Optimization Problems, Models and Methods for Airline Disruption Management from 2010 to 2024". In: *Digital Transportation and Safety* 3.4 (2024), pp. 246–263. DOI: 10.48130/dts-0024-0022.

[45] INCOSE. *Guide to Writing Requirements*. Tech. rep. INCOSE-TP-2010-006-04. Version 4. International Council on Systems Engineering, 2023.

[46] INCOSE. *INCOSE Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities*. 5th. Wiley, 2023. ISBN: 978-1-119-81429-0. URL: https://www.incose.org/about-systems-engineering/system-and-se-definition/systems-engineering-definition (visited on 09/19/2026).

[47] International Civil Aviation Organization. *Global Air Traffic Management Operational Concept*. Doc 9854, AN/458. Montreal, Canada: International Civil Aviation Organization, 2005.

[48] International Civil Aviation Organization. *ICAO Carbon Emissions Calculator Methodology*. Methodology document Version 13.1. International Civil Aviation Organization, Aug. 2024. URL: https://www.icao.int/sites/default/files/environmental-protection/ENVtools/SuportingMaterials/ICEC_Methodology_Passagers_v13_Final.pdf.

[49] International Civil Aviation Organization. *ICAO Aircraft Engine Emissions Databank*. Spreadsheet, hosted by the European Union Aviation Safety Agency. Issue 32. Mar. 2026. URL: https://www.easa.europa.eu/easa-and-you/environment/icao-aircraft-engine-emissions-databank.

[50] ISO/IEC/IEEE. *ISO/IEC/IEEE 15288:2023 Systems and software engineering — System life cycle processes*. Tech. rep. Geneva: International Organization for Standardization, 2023.

[51] Swapnil Sarjerao Jagtap et al. "Model-Based Systems Engineering Approach for a Systematic Design of Aircraft Engine Inlet". In: *AIAA SciTech Forum 2025*. 2025. DOI: 10.2514/6.2025-1410.

[52] Priti Jain. "Personal Knowledge Management: The Foundation of Organisational Knowledge Management". In: *South African Journal of Libraries and Information Science* 77.1 (2011), pp. 1–14. DOI: 10.7553/77-1-62.

[53] Abdessamad Jarrar and Youssef Balouki. "Formal Modeling of a Complex Adaptive Air Traffic Control System". In: *Complex Adaptive Systems Modeling* 6.1 (2018), p. 6. DOI: 10.1186/s40294-018-0056-4.

[54] Heiko Kern, Axel Hummel, and Stefan Kühne. "Towards a Comparative Analysis of Meta-Metamodels". In: *Proceedings of the Compilation of the Co-located Workshops on DSM'11, TMC'11, AGERE! 2011, AOOPES'11, NEAT'11, & VMIL'11 (SPLASH 2011 Workshops)*. ACM, 2011, pp. 7–12. DOI: 10.1145/2095050.2095053.

[55] Barry Kirwan and Mary Flynn. "Identification of Air Traffic Controller Conflict Resolution Strategies for the CORA (Conflict Resolution Assistant) Project". In: *4th USA/Europe Air Traffic Management R&D Seminar*. Santa Fe, NM, Dec. 2001.

[56] Konstantia Kontodimou et al. "Proactive Aircraft Turnaround Buffer Optimization Integrating Machine Learning and Scenario Analysis". In: *Journal of the Air Transport Research Society* 6 (2026), p. 100117. DOI: 10.1016/j.jatrs.2026.100117.

[57] Deepak Kulkarni. *Impact of Slot Substitution on CTOP Performance*. Technical Memorandum NASA/TM-2019-220158. Moffett Field, CA: NASA Ames Research Center, Mar. 2019.

[58] Tianning Liu et al. "Model-Based Systems Engineering Supporting Architecture Modeling of Air Traffic Management System and Model Verifying Based on SMT". In: *Knowledge Science, Engineering and Management (KSEM 2024)*. Vol. 2269. Communications in Computer and Information Science. Springer, 2025, pp. 183–197. DOI: 10.1007/978-981-96-0178-3_13.

[59] Manoj Lohatepanont and Cynthia Barnhart. "Airline Schedule Planning: Integrated Models and Algorithms for Schedule Design and Fleet Assignment". In: *Transportation Science* 38.1 (2004), pp. 19–32. DOI: 10.1287/trsc.1030.0026.

[60] Jing Lu et al. "Harnessing Digital Twin Technology for Enhanced Aircraft Turnaround Efficiency". In: *Transport Economics and Management* 3 (2025), pp. 334–345. DOI: 10.1016/j.team.2025.09.001.

[61] Jinzhi Lu et al. "Design Ontology Supporting Model-based Systems-engineering Formalisms". In: *arXiv preprint* (2020).

[62] Jinzhi Lu et al. "Semantic Model-based Systems Engineering based on KARMA: A Research and Practice Roadmap 2025". In: *32nd Annual INCOSE International Symposium*. Detroit, MI, 2022.

[63] Christopher A. Lupp et al. "A Theoretical Foundation and Practical Demonstration of Integrating MDO, MBSE and the Digital Thread". In: *AIAA SciTech Forum*. Air Force Research Laboratory / University of Dayton Research Institute. 2026.

[64] Apoorv Maheshwari, C. Robert Kenley, and Daniel A. DeLaurentis. "Creating Executable Agent-Based Models Using SysML". In: *25th Annual INCOSE International Symposium (IS2015)*. Bellevue, WA, 2015.

[65] N. A. M. Maiden et al. "Model-Driven Requirements Engineering: Synchronising Models in an Air Traffic Management Case Study". In: *16th International Conference on Advanced Information Systems Engineering (CAiSE 2004)*. Vol. 3084. Lecture Notes in Computer Science. Riga, Latvia: Springer, 2004, pp. 368–383.

[66] Tom McDermott et al. "AI4SE and SE4AI: A Research Roadmap". In: *INSIGHT* 23.1 (2020), pp. 8–14. DOI: 10.1002/inst.12278.

[67] Faida Mhenni et al. "Flight Control System Modeling with SysML to Support Validation, Qualification and Certification". In: *13th IFAC and IEEE Conference on Programmable Devices and Embedded Systems*. Vol. 49-3. IFAC-PapersOnLine. Elsevier, 2016, pp. 453–458. DOI: 10.1016/j.ifacol.2016.07.076.

[68] Yaniv Mordecai. "Conceptual Modeling of Cyber-Physical Gaps in Air Traffic Control". In: *Complex Adaptive Systems Conference with Theme: Cyber Physical Systems and Deep Learning (CAS 2018)*. Vol. 140. Procedia Computer Science. Elsevier, 2018, pp. 21–28. DOI: 10.1016/j.procs.2018.10.288.

[69] Ryota Mori. "Aircraft Mass Estimation Using Cruise Flight Profile". In: *33rd Congress of the International Council of the Aeronautical Sciences (ICAS)*. Stockholm, Sweden, 2022. URL: https://www.icas.org/icas_archive/ICAS2022/data/papers/ICAS2022_0105_paper.pdf.

[70] Sai V. Mudumba et al. "Model-Based Systems Engineering Approach for Simulating Urban Air Mobility Maturity Level 5 (UML-5) Operations". In: *AIAA AVIATION Forum*. 2022. DOI: 10.2514/6.2022-4075.

[71] Pamela Munro and Richard Mogford. "Managing Variability: A Cognitive Ethnography of the Work of Airline Dispatchers". In: *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*. Vol. 62. 1. 2018, pp. 182–186. DOI: 10.1177/1541931218621043.

[72] Nadine Pilon, Laurent Guichard, and Katherine Cliff. "Reducing Impact of Delays using Airspace User-Driven Flight Prioritisation: User Driven Prioritisation Process Validation Simulation and Results". In: *9th SESAR Innovation Days*. Paper 39. ISSN 0770-1268. Dec. 2019. URL: https://sesarju.eu/sites/default/files/documents/sid/2019/papers/SIDs_2019_paper_39-1.pdf.

[73] Esa M. Rantanen and Christopher D. Wickens. "Conflict Resolution Maneuvers in Air Traffic Control: Investigation of Operational Data". In: *The International Journal of Aviation Psychology* 22.3 (2012), pp. 266–281. DOI: 10.1080/10508414.2012.691048.

[74] Bill Roberson. "Fuel Conservation Strategies: Cost Index Explained". In: *Boeing AERO* QTR_02 (2007), pp. 26–28. URL: https://skybrary.aero/sites/default/files/bookshelf/1956.pdf.

[75] Ítalo Romani de Oliveira et al. "A Predictive Services Architecture for Efficient Airspace Operations". In: *2026 Integrated Communications, Navigation and Surveillance Conference (ICNS)*. IEEE. 2026. DOI: 10.1109/ICNS69853.2026.11570268.

[76] Ahmed R. Sadik et al. "Urban Air Mobility as a System of Systems: An LLM-Enhanced Holonic Approach". In: *20th Annual System of Systems Engineering Conference (SOSE)*. IEEE. 2025.

[77] Mateus Santana et al. "The Aircraft Recovery Problem: A Systematic Literature Review". In: *EURO Journal on Transportation and Logistics* 12 (2023), p. 100117. DOI: 10.1016/j.ejtl.2023.100117.

[78] Michael Schmidt. "A Review of Aircraft Turnaround Operations and Simulations". In: *Progress in Aerospace Sciences* 92 (2017), pp. 25–38. DOI: 10.1016/j.paerosci.2017.05.002.

[79] Charles A. Schultz, David Thipphavong, and Heinz Erzberger. "Adaptive Trajectory Prediction Algorithm for Climbing Flights". In: *AIAA Guidance, Navigation, and Control Conference*. AIAA-2012-4931. Venue and paper number unconfirmed from the PDF; DOI not found. 2012. URL: https://ntrs.nasa.gov/api/citations/20140004896/downloads/20140004896.pdf.

[80] Florian Schummer and Maximillian Hyba. "An Approach for System Analysis with Model-based Systems Engineering and Graph Data Engineering". In: *Data-Centric Engineering* 3 (2022), e33. DOI: 10.1017/dce.2022.33.

[81] Thomas L. Seamster, Eric Chevalley, and Barbara G. Kanki. *Collaborative Systems Assessment: Flightdeck, Air Traffic Control, Flight Operations Center and Automation*. Draft Final Report, FAA task 05-04 / AJP61SSP-0050. NASA Ames Research Center, Aug. 2011.

[82] SESAR Joint Undertaking. *SESAR Solutions Catalogue 2019*. Tech. rep. Catalogue number MG-07-18-083-EN-N. Luxembourg: Publications Office of the European Union, 2019. DOI: 10.2829/998701.

[83] SESAR Joint Undertaking. *European ATM Master Plan: Making Europe the Most Efficient and Environmentally Friendly Sky to Fly in the World*. Tech. rep. SESAR Joint Undertaking, 2025.

[84] SESAR Joint Undertaking. *SESAR eATM ATM Capability Model (Extracted Capability Catalogue)*. 2025.

[85] Kapil Sheth and Sebastian Gutierrez-Nolasco. "Analysis of Factors for Incorporating User Preferences in Air Traffic Management: A System Perspective". In: *27th International Congress of the Aeronautical Sciences (ICAS 2010)*. Nice, France, 2010.

[86] Kapil S. Sheth et al. "Simulations of Credits Concept with User Input for Collaborative Air Traffic Management". In: *AIAA Guidance, Navigation, and Control Conference*. AIAA 2010-8079. Toronto, Ontario, Canada, Aug. 2010. DOI: 10.2514/6.2010-8079.

[87] Sonali Sinha Roy et al. "From Ontology to System Architecture: An MBSE Approach Toward the Realization of Urban Air Mobility". In: *34th Congress of the International Council of the Aeronautical Sciences (ICAS)*. Florence, Italy, 2024.

[88] Junzi Sun, Jacco M. Hoekstra, and Joost Ellerbroek. "OpenAP: An Open-Source Aircraft Performance Model for Air Transportation Studies and Simulations". In: *Aerospace* 7.8 (2020), p. 104. DOI: 10.3390/aerospace7080104. URL: https://openap.dev (visited on 10/10/2026).

[89] Serkan Turhan, Fatma Gzara, and Samir Elhedhli. "Workload Balancing for Flight Dispatchers". In: *Computers & Operations Research* 186 (2026), p. 107303. DOI: 10.1016/j.cor.2025.107303.

[90] Sabine Wilke, Arnab Majumdar, and Washington Y. Ochieng. "Airport Surface Operations: A Holistic Framework for Operations Modeling and Risk Management". In: *Safety Science* 63 (2014), pp. 18–33. DOI: 10.1016/j.ssci.2013.10.015.

[91] Tim Wittenborg et al. *Knowledge-Based Aerospace Engineering: A Systematic Literature Review*. 2025.

[92] Yifan Xu, Sebastian Wandelt, and Xiaoqian Sun. "Airline Scheduling Optimization: Literature Review and a Discussion of Modelling Methodologies". In: *Intelligent Transportation Infrastructure* 2 (2024), liad026. DOI: 10.1093/itinfr/liad026.

[93] Anzhuo Yao et al. "System-of-Systems Safety for Low-Altitude Aviation Transportation". In: *Reliability Engineering and System Safety* 273 (2026), p. 112276. DOI: 10.1016/j.ress.2026.112276.

[94] Haytham Younus et al. "AI- and Ontology-Based Enhancements to FMEA for Advanced Systems Engineering: Current Developments and Future Directions". In: *Applied Sciences* 16.5 (2026), p. 2464. DOI: 10.3390/app16052464.

A Appendices

Plan ref: to-do-list.md S5, S7, S10, S11 (general) — Use appendices for material too extensive for the main body or of interest to only some readers. Label each lettered (Appendix A, B, ...), and ensure every appendix is referenced from somewhere in the main body.

A.1 Full Nominal-Flight ConOps

Plan ref: to-do-list.md S5 — Full strategic-planning through postflight/continuation activity breakdown (see main-body summary, Section 3.4). The phase activity diagrams are placed below (Fly Enroute is in the body, Figure 6); add the text that introduces them and cite each one from it.

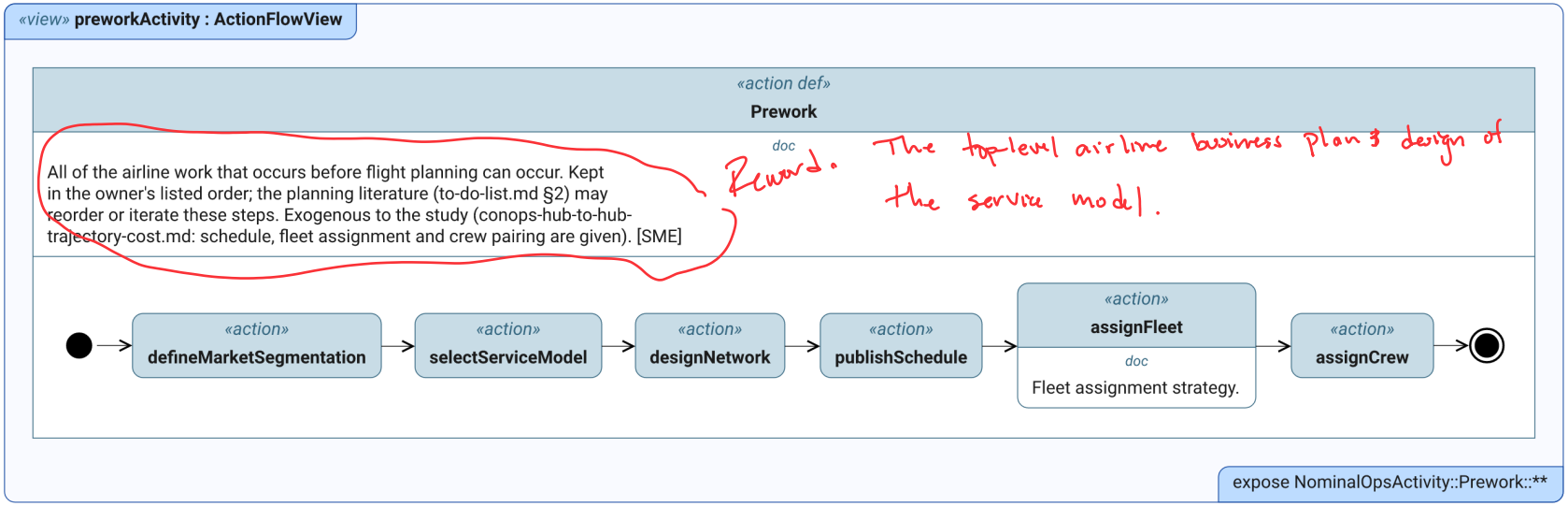


Figure 15: Nominal operations, Prework: airline planning before flight planning. View `preworkActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-18.

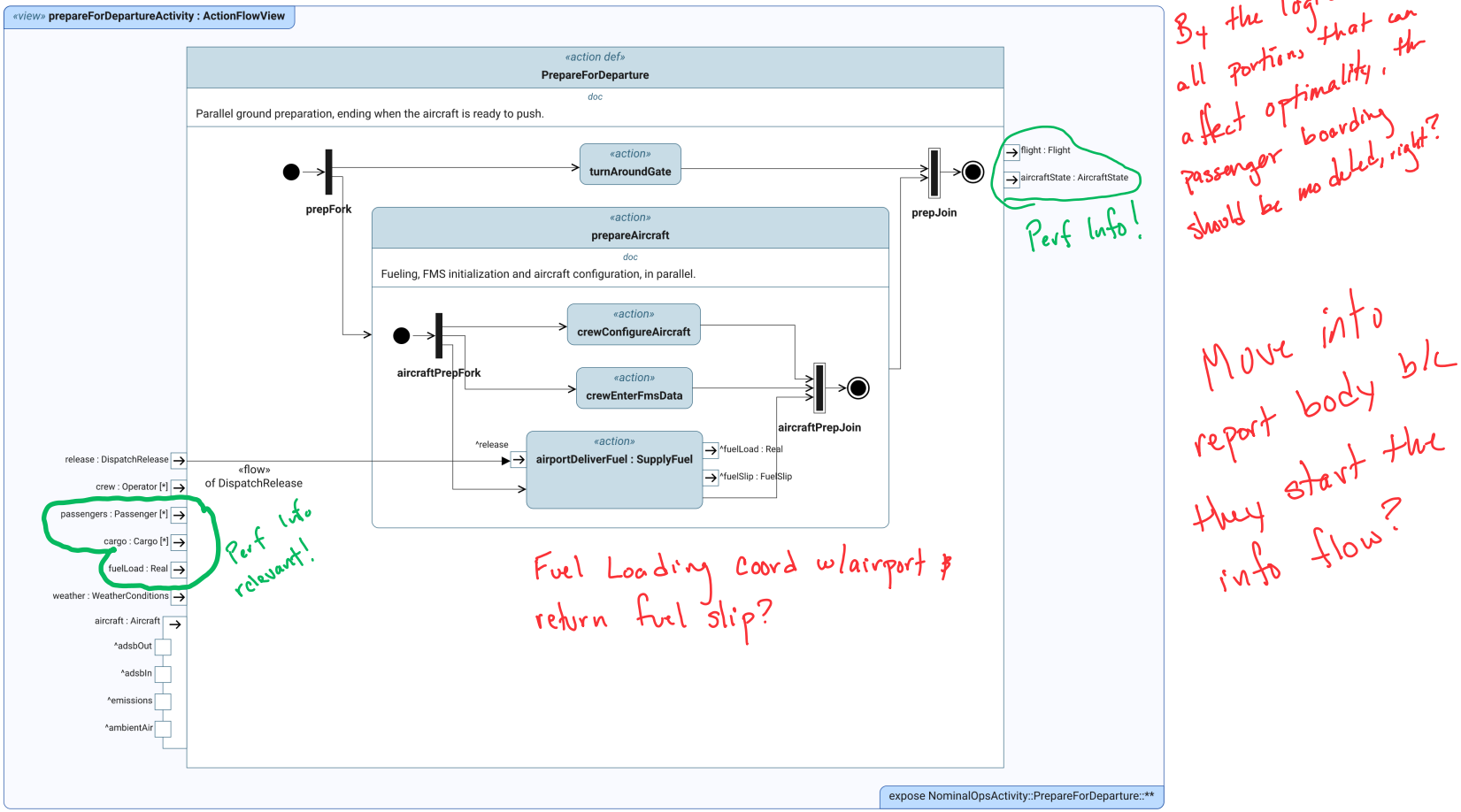


Figure 16: Nominal operations, Prepare for Departure: parallel fueling, FMS entry, configuration and gate turnaround. View `prepareForDepartureActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-18.

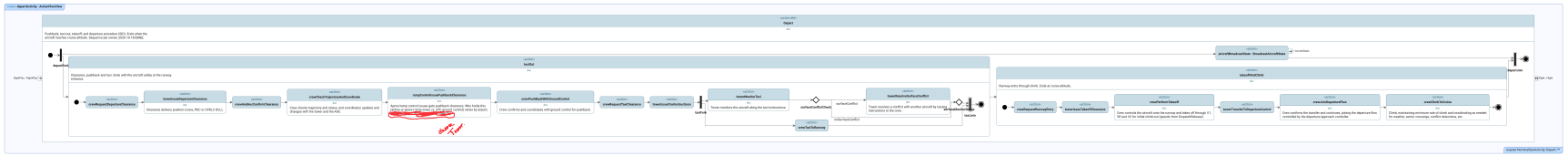


Figure 17: Nominal operations, Depart: clearance, pushback, taxi-out, takeoff and climb. View `departActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-18.



Figure 18: Nominal operations, Arrive: arrival planning, descent, approach, landing and taxi-in. View `arriveActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-18.

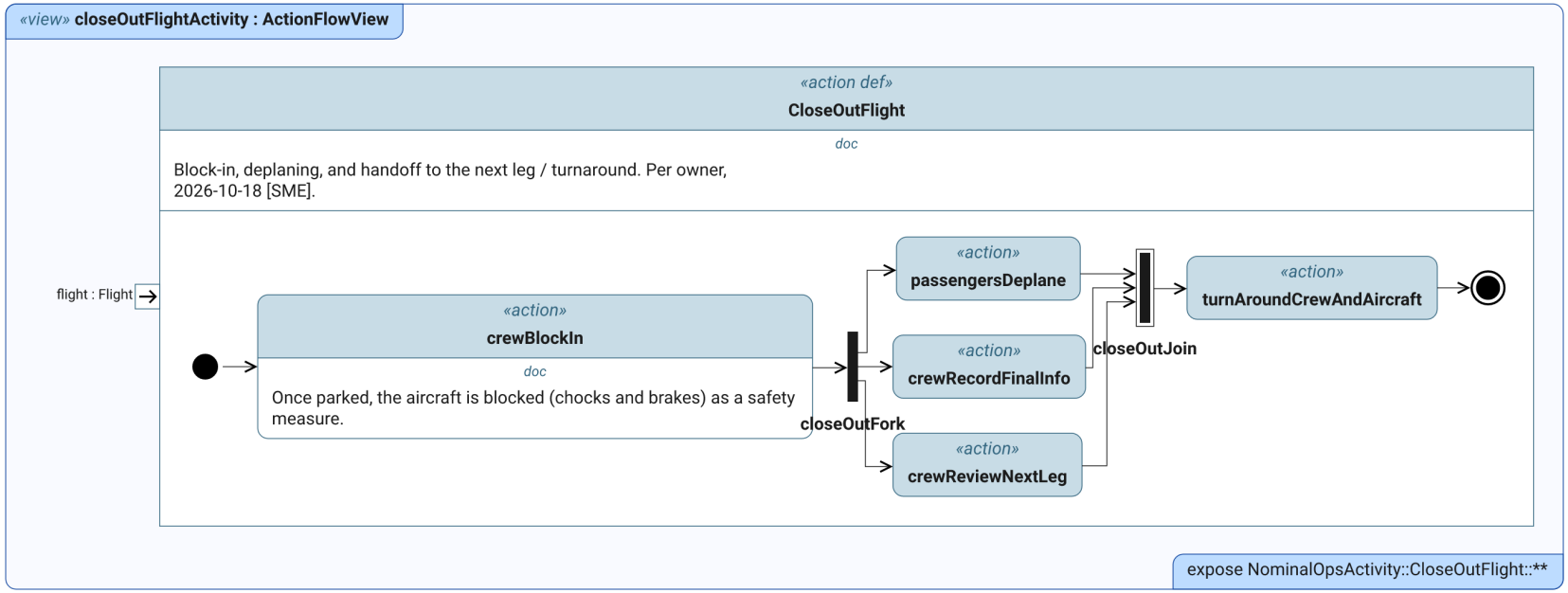


Figure 19: Nominal operations, Close Out Flight: block-in, deplaning and turnaround. View `closeOutFlightActivity`; source `nas_activity_diagram-nominal_ops.sysml`; rendered 2026-10-18.

A.2 Full Stakeholder Register and RACCI Matrix

Plan ref: to-do-list.md S7 — Complete stakeholder register (needs, goals, responsibilities, authorities, resources, constraints, information needs, measures of value) and the full RACCI matrix around operational decisions.

AI-DRAFTED, owner to rewrite — Excerpt of the stakeholder assessment. Ten of the workbook's 48 rows and four of its thirteen columns, chosen by Claude to show each recommended treatment. Choose your own rows, or replace the excerpt with the whole sheet.

Table 7: Excerpt of the stakeholder assessment (`capstone_stakeholder_assessment.xlsx`): ten of 48 candidates. Score is importance (high 3, medium 2, low 1) plus trajectory-decision impact (yes 2, indirect 1, no 0).

Stakeholder (PESTLE domain)	Trajectory-decision impact	Standalone importance	Score	Recommended treatment
Airlines (economic)	Yes	High	5	Standalone
Airports (economic)	Yes	High	5	Standalone
Air traffic controllers (technical)	Yes	High	5	Standalone
Dispatchers (technical)	Yes	High	5	Standalone
Flight crews (social)	Indirect	High	4	Consider / consolidate
Local communities (environmental)	Indirect	High	4	Consider / consolidate
Federal Aviation Administration (political)	No	High	3	Consider / consolidate
Passengers (economic)	No	High	3	Consider / consolidate
Aircraft manufacturers (economic)	Indirect	Low	2	Boundary input / omit
Congress and legislatures (political)	No	Low	1	Boundary input / omit

A.3 Full SysML Diagram Set

Plan ref: to-do-list.md S10 — Diagrams not embedded in the body: full BDD/IBD set, requirements model, and complete traceability matrices (needs → objectives → requirements → systems → activities → information exchanges → aircraft behavior). Not yet modeled as views: BDDs and the requirements model.

Every figure of the model in this report is a view defined in `nas_package_my_views.sysml` and rendered from the SysML v2 text with Syside. Its caption names the view, the `.sysml` file that holds the content, and the date of the render. As of this draft all five views appear in the body: Figure 4 (Section 3.3), Figure 5 (Section 3.4), and Figures 7, 8 and 9 (Section 3.5). Views added to the model later appear below.

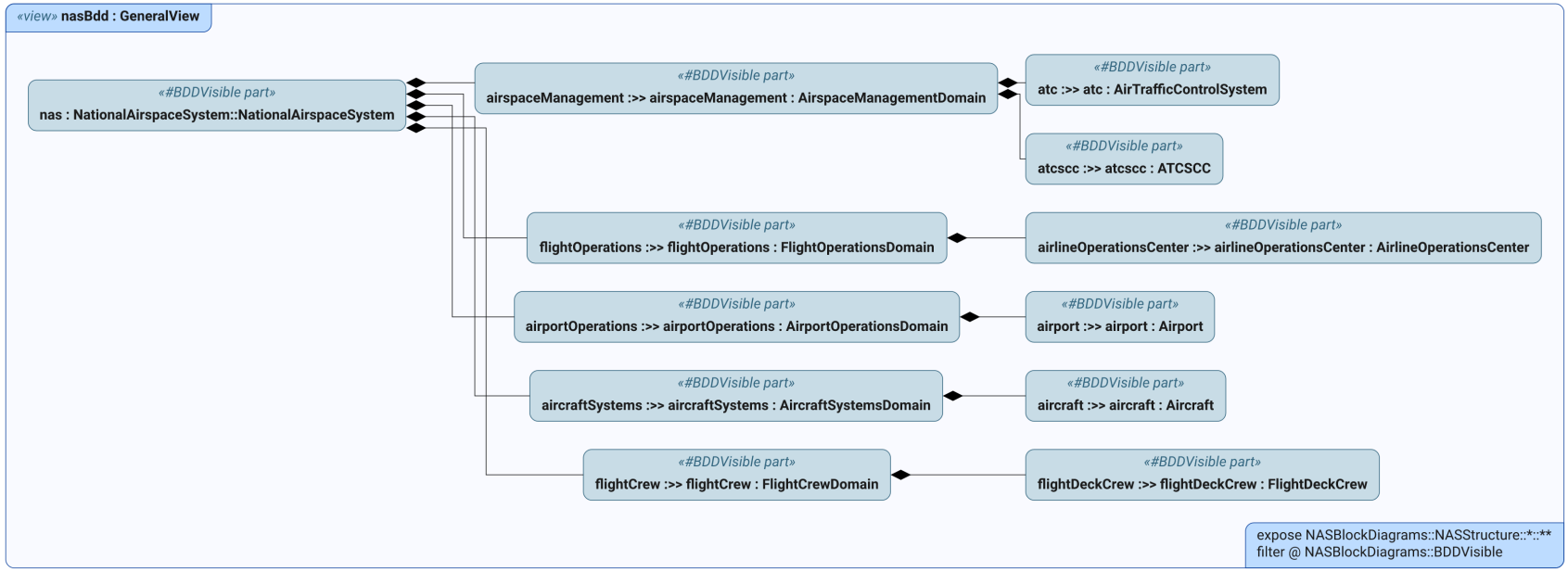


Figure 20: NAS block definition diagram: the modeled domains and the context-diagram systems, composition only. View `nasBdd`; source `nas_block_diagrams.sysml`; rendered 2026-10-19.

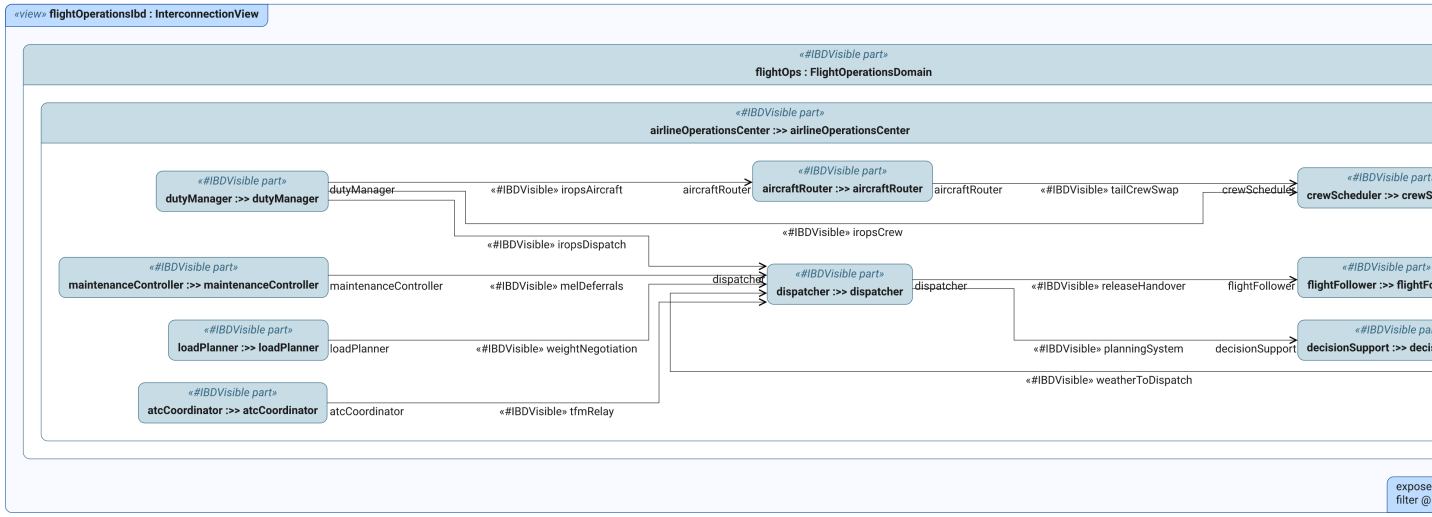


Figure 21: Flight operations internal block diagram: the airline operations center's roles. View `flightOp` `nas_block_diagrams.sysml`; rendered 2026-10-19.

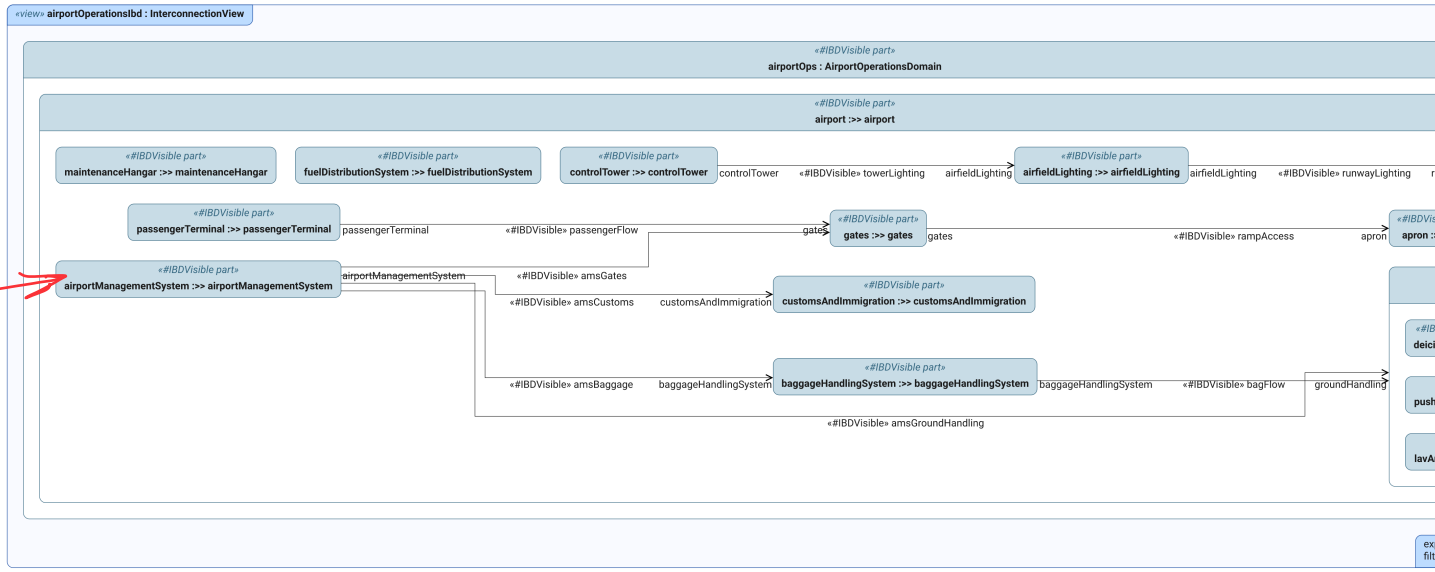


Figure 22: Airport operations internal block diagram. View `airportOperationsIbd`; source `nas_block_diagrams` 10-19.



Figure 23: Flight crew internal block diagram. View `flightCrewIbd`; source `nas_block_diagrams.sysml`; re

A.4 Full Myopic-Optimization Conflict Catalog

Plan ref: to-do-list.md S9 — All candidate conflicts not carried into the main-body discussion or the S11–S14 demonstration scenario, each with: decision maker, decision variable, local objective/constraints, affected stakeholders, externalized costs, SoS consequence, and information needed to recognize the consequence.

A.5 Optimization Experiment Detail

Plan ref: to-do-list.md S11, S12 — Full experiment matrix, constituent-model validation notes, and any sensitivity-analysis detail not included in Section 5.

Plan ref: to-do-list.md S11 — Full options table: for each of the nine options, the maneuver per aircraft, instructions now, follow-up later, and what the decider must know to use it. Draft in `knowledge/models/experiment-options-and-scoring.md` §2.

Plan ref: to-do-list.md S11, S12 — Every option scored at four levels (the table from which each case picks one row), the per-viewpoint cost statements with their symbols defined, and the table of abstractions with reason and risk.

A.6 Generative AI Use Disclosure

REQUIRED FOR INTERIM REPORT #2 (due 2026-10-25) — Generative AI disclosure: owner to draft. Required by the course policy whenever GenAI is used; Interim Report #2 is the first submission after the policy was read (2026-09-21).

A.7 Source Register

Plan ref: to-do-list.md S1–S4 — Reference or reproduce the evidence/source register (`evidence/source-register.md`): source, publication type, NAS lifecycle phase, systems/stakeholders addressed, architectural and optimization evidence extracted, and confidence/limitations.